

Benchmarking Deep Reinforcement Learning for Cartesian Path Tracking of Free-Floating Space Robot Manipulators

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ABSTRACT Free-floating space robots exhibit a strong dynamic coupling between manipulator motion and base attitude, which complicates Cartesian path tracking of the end-effector. Labeled on-orbit data for supervised learning are scarce. This work therefore benchmarks six reinforcement learning (RL) algorithms (TRPO, TD3, SAC, PPO, PG, and DDPG) for Cartesian path tracking of a free-floating space robot in a physics-based Simulink/Simscape environment with runtime safety monitoring. Each configuration is trained with ten seeds. The success rate and the performance of the successful runs are evaluated separately with unified key performance indicators. With default hyperparameters, only TRPO completes all ten runs without a safety abort, whereas the other agents terminate 40% to 80% of their evaluation episodes. Even in their successful runs, SAC, DDPG, and TD3 track the path with a significantly larger error than TRPO. A tuned PPO configuration also completes all runs and reaches the lowest mean squared tracking error (0.0041 compared with 0.0145 for TRPO) with a shorter training time, but this difference is not statistically significant. On a piecewise-linear path, the tuned agent tracks significantly better than TRPO. Both agents complete all runs under observation noise, reduced torque limits, an external joint torque, and model parameters increased by 50%. All results are obtained in simulation under controlled assumptions and do not establish operational readiness. The safety component comprises runtime monitors for selected properties and an offline torque-bound check on a simplified model, not a formal proof of closed-loop safety.

INDEX TERMS Free-floating on-orbit servicing, Proximal Policy Optimization (PPO), reinforcement learning, safety monitoring, space robotics, trajectory tracking

I. INTRODUCTION AND BACKGROUND

Free-floating robot manipulators operating in space for satellite inspection, capture, or de-orbiting introduce control challenges not encountered with terrestrial robots [1]. In micro-gravity, the robot's base (typically a spacecraft) is not fixed, and the thrusters are inactive. Manipulator motions therefore change the position and orientation of the base, which in turn modifies the pose of the end-effector. This reaction is a consequence of the conservation of linear and angular momentum [2]. Operating the manipulator under free-floating dynamics avoids the consumption of propellant (e.g., hydrazine, N_2H_4) and the excitation of structural modes by the pulse-width modulation of thruster control at certain frequencies [3]. This can extend the operational lifetime of the spacecraft.

However, a free-floating base complicates manipulator control. Standard motion-planning approaches developed for fixed-base robots can fail to drive the end-effector to the

desired target. Additional constraints, such as communication latency and limited onboard computational resources, further complicate continuous real-time control and base stabilization [1], [4]. Model-based approaches that assume an accurate dynamic model (e.g., reactionless planning or precise attitude control) are sensitive to model uncertainty and unmodeled dynamics. Supervised data-driven approaches are also difficult to apply because labeled training data of sufficient amount and quality are typically unavailable in the space sector. In addition, tracking capability under a free-floating base often has to be demonstrated before the physical system is assembled and deployed [5]–[7].

These challenges motivate the development and simulation of AI-enabled digital twins (DTs) [8], as illustrated in Fig. 1. Unlike standard DTs that primarily mirror sensed real-time data from physical counterparts, the DTs considered here combine physics-based simulation with learning components

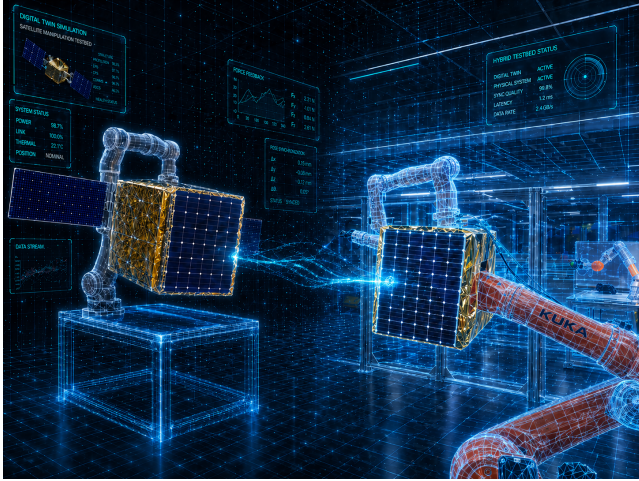


FIGURE 1: Virtual testbeds based on AI-driven digital twins. They allow learning-based control methods for in-space operations to be developed under repeatable and controllable conditions. Illustration created using a picture of the physical setup and with the help of GPT-Image-2.

to contextualize system states and evaluate representative operating conditions. These conditions include uncertain dynamics, disturbances, faults, and contact-rich interactions associated with satellite capture. Faster-than-real-time simulation and systematic ablation studies can support controller design and help assess sim-to-real effects. In this role, AI-enabled DTs can provide a training and validation environment for learning-based controllers, in particular when flight data are limited and the constraints are safety-critical.

Reinforcement learning (RL) has been increasingly applied in robotics. An RL agent learns a control policy through trial-and-error interaction with its environment. Recent advances in deep RL have addressed a range of complex continuous-control tasks, although applications to space robots remain comparatively recent [9], [12]–[14], [16].

This work investigates continuous-action deep RL for Cartesian path tracking of the end-effector of a free-floating space manipulator. The contribution differs from prior work in three respects. First, six RL algorithms are compared under the same conditions (see Section VI), complemented by ablation studies (see Section VII-B). The works listed in Table 1 do not provide such a comparison. Second, selected constraints such as collisions and joint limits are monitored at runtime with formal-methods-based tools integrated into the simulation toolchain (see Section V). Third, base attitude minimization during arm motion is treated as an explicit control objective in conjunction with the constraint monitoring above. The induced base disturbance is not addressed in most of the works listed in Table 1. To the authors' knowledge, a unified comparison of RL algorithms for Cartesian path tracking of free-floating space robots that jointly considers base attitude minimization and formal-methods-based constraint monitoring remains unavailable.

As shown in Table 1, the surveyed prior contributions

typically evaluate a single RL algorithm on a specific subtask, which limits conclusions about the relative merits of different learning algorithms for free-floating space robot control. In addition, the surveyed works do not combine three aspects. These are (i) a multi-algorithm benchmark under unified conditions, (ii) explicit minimization of the base attitude disturbance alongside end-effector tracking, and (iii) formal-methods-based runtime safety monitoring in the RL training and evaluation loop. This work addresses these aspects together in a reproducible simulation benchmark with runtime monitoring of selected safety properties, which extends the individual studies listed in Table 1.

The main contributions of this work are summarized as follows:

- *Comparative evaluation of RL agents.* Six continuous-action RL algorithms are implemented and evaluated on the same path-tracking task. The algorithms are Trust Region Policy Optimization (TRPO) [17], Twin Delayed Deep Deterministic Policy Gradient (TD3) [18], Soft Actor-Critic (SAC) [19], Proximal Policy Optimization (PPO) [20], vanilla Policy Gradient (PG) [21], and Deep Deterministic Policy Gradient (DDPG) [43]. The comparison is performed under unified conditions with ten seeds per algorithm. It reports separately how often training succeeds and how well the successful runs perform.
- *Integration of formal-methods-based safety monitoring.* Formal-methods tooling is integrated into the learning loop [2], [23]–[25]. A collision monitor, a joint-limit monitor, and a momentum-conservation check run at runtime, and the joint-torque bound is additionally checked offline with Simulink Design Verifier on a simplified abstraction of the policy. This combination monitors selected safety properties (collision events, joint limits, and bounded actuator commands) during training and testing. It does not constitute a formal proof of closed-loop stability, task completion, or collision avoidance for the full nonlinear multibody model.
- *Simulation environment for multi-criteria evaluation in on-orbit servicing.* A MATLAB/Simulink simulation framework is developed for a free-floating space robot. It uses the Simscape physics engine and a multibody model derived from a Unified Robot Description Format (URDF) file. The robot consists of a cuboid base and a 4-degree-of-freedom (4-DOF) revolute arm, modeled with consistent kinematic, inertial, and sensor/actuator parameters. The associated RL environment provides observations and rewards through a composite reward function that penalizes end-effector tracking error, base motion, large joint velocities, and excessive control effort. Together with the reset conditions, this reward shaping allows the agent to learn the task while penalizing base disturbance and violations of the monitored constraints [22], [26]–[29].
- *Optimization of the PPO agent.* The default PPO agent

TABLE 1: Recent advances in deep RL-based motion management for free-floating space robots. Abbreviations: RL Alg. = reinforcement learning algorithm; DOF = degrees of freedom; Obs./Act. Dim. = observation-/action-space dimension; vel. = joint velocities; rates = joint rate commands; pos./att. = position/attitude; pos./ori. = position/orientation; EE = end-effector; Formal Meth. = formal-methods-based safety components; Base Dist. Min. = base disturbance minimization; d_{safe} = minimum admissible distance; τ_{max} = torque bound; K_1 – K_9 = evaluation KPIs (see Section IV); N/R = not reported; [†]runtime monitors for selected properties and an offline torque-bound check on a simplified model, not a formal proof of closed-loop safety.

Contribution	RL Alg.	Task Objective	DOF	Simulator	Obs./Act. Dim.	Safety Limits	Eval. Metrics	Formal Meth./ Base Dist. Min.
[9], 2023	SAC	Obstacle avoidance	7	OpenAI Gym	14 / 7D vel.	Reward-based	Success rate	No/No
[10], 2024	DDPG	Collision avoidance	6	Custom	N/R / 6D vel.	Reward-based	Success rate	No/No
[11], 2024	PPO	Joint failure adaptation	7	MATLAB Simscape + SPART	32 / 7D rates	Pos./att. threshold	Per-joint success rate	No/No
[12], 2024	DDPG	Path planning	6	Custom	32 / 6D vel.	Reward-based	Convergence, task completion	No/No
[13], 2025	PPO	Trajectory planning	6	PyBullet	N/R / 6D	Joint limits	Success rate, pos./ori. error	No/No
[14], 2025	LSAC	Trajectory post-processing	N/R	Custom + air-bearing	N/R / N/R	Uncertainty bounds	Tracking error	No/No
[15], 2025	DDPG	Trajectory planning	N/R	Custom	N/R / N/R	Base dist. (soft)	Reward, base dist. red.	No/Yes
Ours	PG, PPO, TRPO, DDPG, TD3, SAC	Cartesian EE tracking	4	MATLAB/Simulink + Simscape	23/4D τ	$d_{\text{safe}}, \tau_{\text{max}},$ joint limits	K_1 – K_9	Partial[†]/Yes

converged in only three of ten runs, but its successful runs tracked the path with an error similar to that of TRPO. PPO is therefore tuned to reduce this seed dependence. Hyperparameters such as the learning rates, the rollout horizon, the mini-batch size, and the entropy weight are adjusted. The hidden layers of the default networks are kept. The tuned agent completes all ten runs without an observed violation of the monitored properties and reaches the lowest mean tracking error in the benchmark.

Within the considered simulation setup, TRPO and the tuned PPO agent completed all runs without observed violations of the monitored constraints. The tuned policy also reduced the tracking error and the base disturbance relative to the successful runs of the default PPO configuration. The remainder of this work is organized as follows. Section III describes the system model and the simulation environment. Section IV introduces the reinforcement learning framework. Section V presents the safety monitors and formal-methods components. Section VI reports the comparative evaluation of the six RL algorithms. Section VII describes the optimization of the PPO agent and its evaluation on circular and piecewise-linear references. Section VIII discusses the results and concludes the work.

II. RELATED WORK

A. CLASSICAL CONTROL TECHNIQUES FOR SPACE ROBOTS

Model-based approaches have historically dominated space robot control. Computed torque control and resolved-motion rate methods use accurate dynamic models to achieve precise end-effector positioning [1]. Reactionless path planning explicitly minimizes base disturbance. It exploits the robot's

redundancy and a dynamics-related null-space to design the joint velocities. Attitude regulation methods instead use reaction wheels or thrusters to counteract the momentum transfer [2], [7]. These approaches offer strong performance guarantees and interpretability when the system model is accurate. However, they are sensitive to model uncertainties and unmodeled dynamics and require a detailed model identification [37]. They are therefore difficult to apply under the data-scarce and hazardous conditions of on-orbit operations.

B. REINFORCEMENT LEARNING-BASED CONTROL TECHNIQUES

RL has been applied successfully to various robotic manipulation tasks [22], [30]. Policy gradient methods (PG, PPO, TRPO) and actor-critic algorithms (DDPG, TD3, SAC) have been widely applied to continuous control problems [17]–[21]. Applied to free-floating space robots, RL-based methods have been investigated for path planning [12], [16], obstacle avoidance [9], trajectory tracking [13], and failure management [11]. RL does not require an explicit dynamic model of the system, which is relevant when model parameters are uncertain or unavailable. However, current RL-based approaches share recurring limitations. Most studies evaluate only a single algorithm, address a single objective, omit base attitude minimization, and do not provide systematic benchmarking under comparable conditions.

C. HYBRID SAFETY-AWARE CONTROL SYSTEMS

Safe RL integrates safety constraints through formal verification, barrier functions, and constrained optimization [24], [25], [31], [32]. Shielding approaches [25] intercept unsafe actions at runtime and substitute corrective commands, while control barrier functions (CBFs) [32] encode safety as a continuously enforced constraint on the policy's action set. These

hybrid methods combine the adaptability of RL with formally specified runtime constraints. They help to enforce properties such as joint-limit compliance and collision avoidance. However, they have rarely been applied to free-floating space robots. Existing works seldom integrate formal-methods-based safety monitoring into the RL training loop or combine it with base attitude minimization objectives.

D. NEED FOR BENCHMARK STUDIES

Comparative evaluations of RL algorithms for space robot control are scarce. Table 1 surveys recent deep RL contributions to free-floating space robot motion management. As shown, all surveyed works address only individual goals with a single algorithm. None simultaneously combines multi-algorithm and multi-criteria benchmarking, base disturbance minimization, and formal-methods-based safety monitoring. This absence of unified evaluation makes it difficult to draw conclusions about relative algorithm performance and to guide algorithm selection for future missions.

E. RESEARCH GAP

The survey above points to three gaps in the existing literature. First, none of the surveyed works compares multiple RL algorithms under identical conditions for Cartesian end-effector tracking of free-floating space robots. Second, base attitude minimization, a safety-critical requirement in on-orbit operations, is omitted in most RL-based contributions (see Table 1). Third, the integration of formal-methods-based safety monitoring into the RL training and evaluation loop has not been reported for this class of systems in the surveyed works. This work addresses the three gaps together within a controlled simulation benchmark. It benchmarks six RL algorithms under unified conditions and treats base attitude minimization as an explicit control objective. Runtime monitoring of selected safety properties is integrated into the RL training and evaluation loop. The actuator-torque bound is additionally checked offline on a simplified verification model.

III. SYSTEM MODELING AND SIMULATION ENVIRONMENT

A. ADVANTAGES OF AI-ENDOWED DIGITAL TWIN TECHNOLOGY IN ON-ORBIT SERVICING

AI-driven DTs have been applied in on-orbit servicing, for example for the automated lifecycle management of satellites and for the orchestration of agents that carry out assembly, integration, and testing tasks [33], [34]. DTs combine contextualized real-time sensor data with AI methods that account for uncertainty. This allows simulation-based analysis under various operational conditions at reduced cost. Verification and validation can also benefit, because changes in system design or operational scenarios can be described in scenario spaces and linked to DTs [35].

In this work, the DTs of the target satellite and the chaser space robot are based on an information model. It provides

TABLE 2: Parameters of the simulated free-floating space robot.

Parameter	Value
Base, cube with 1 m edge length, mass	30 kg
Base, principal moments of inertia	6 kg m ²
Links (4), length and mass	1 m, 2 kg
Links, principal moments of inertia	0.2 kg m ²
Viscous joint damping	1.5 N m s/rad
Joint limits q_1 and q_2 to q_4	$\pm 85^\circ$ and $\pm 170^\circ$
Joint-limit stiffness and damping	3000 N m/rad, 40 N m s/rad
Torque limit $\tau_{\max}$	2 N m
Gravity	$[0 \ 0 \ 0]^T$

a structured and interoperable description of both systems, including their kinematics, inertial properties, joint configurations, and physical constraints. The AI capability is provided by the deep RL agents, which are trained and evaluated through interaction with the corresponding DTs and with additional DTs of representative in-space operation scenarios. This distinguishes the approach from classical simulation with pre-specified controllers. The robotics-related DT hosts a learning-based agent that derives its control strategy from experience. The DTs are therefore used as training and validation environments for learned control strategies. During training, the initial joint configuration of each episode is sampled from a bounded uniform distribution. The agent therefore learns from a range of admissible initial conditions instead of a single fixed start.

B. ROBOT MODEL

The virtual testbed simulates a free-floating space manipulator derived from the Spacecraft Robotics Toolkit (SPART) example [36]. The robot consists of a cuboid base (serving as the spacecraft) and a 4-DOF serial manipulator (see Fig. 2). The arm has four revolute joints and four links. Its kinematic structure and inertial properties are specified in a URDF file (`SpaceRobot.urdf`) imported into Simulink's multibody environment [26]–[28]. The geometry and the inertial parameters of the base and the links are listed in Table 2. The same values are used in the URDF file and in the Simscape model.

The resulting multibody model captures the dynamic coupling between the floating base and the arm [2].

C. DYNAMIC SIMULATION IN MATLAB/SIMULINK

The simulation is implemented in Simulink using the Simscape Multibody physics engine. Gravity is set to $[0 \ 0 \ 0]^T$ to emulate the microgravity orbital environment. The base is connected to the inertial world frame through an unactuated 6-DOF joint [38], allowing free translation and rotation. No external forces or torques act on the system, so the base moves only in response to the internal torques generated by the manipulator joints. Under this configuration, the total linear and angular momentum are conserved, and arm motions induce a reactive motion of the base [2], [7], [29], [32], [38].

The simulation is based on idealizing assumptions that are common in RL-based space robotics research. They are stated here explicitly to support reproducibility and a fair

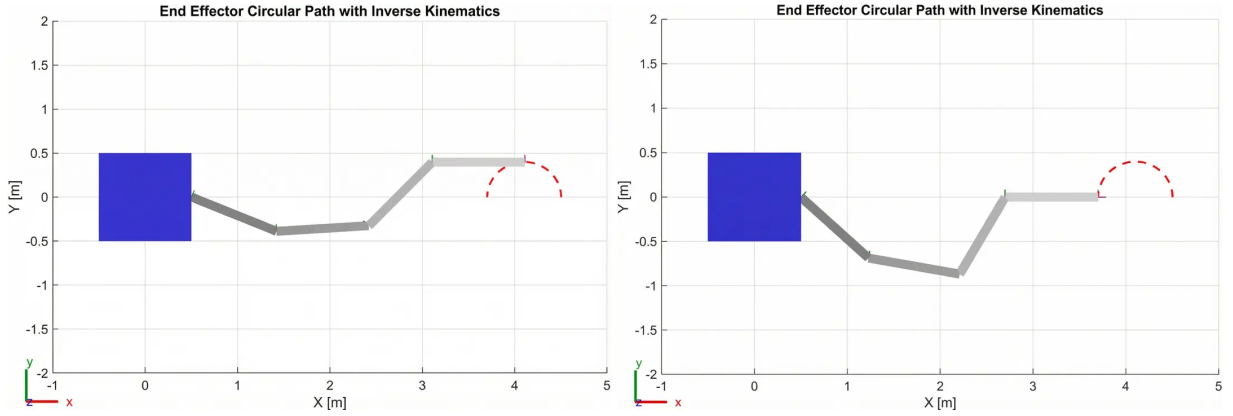
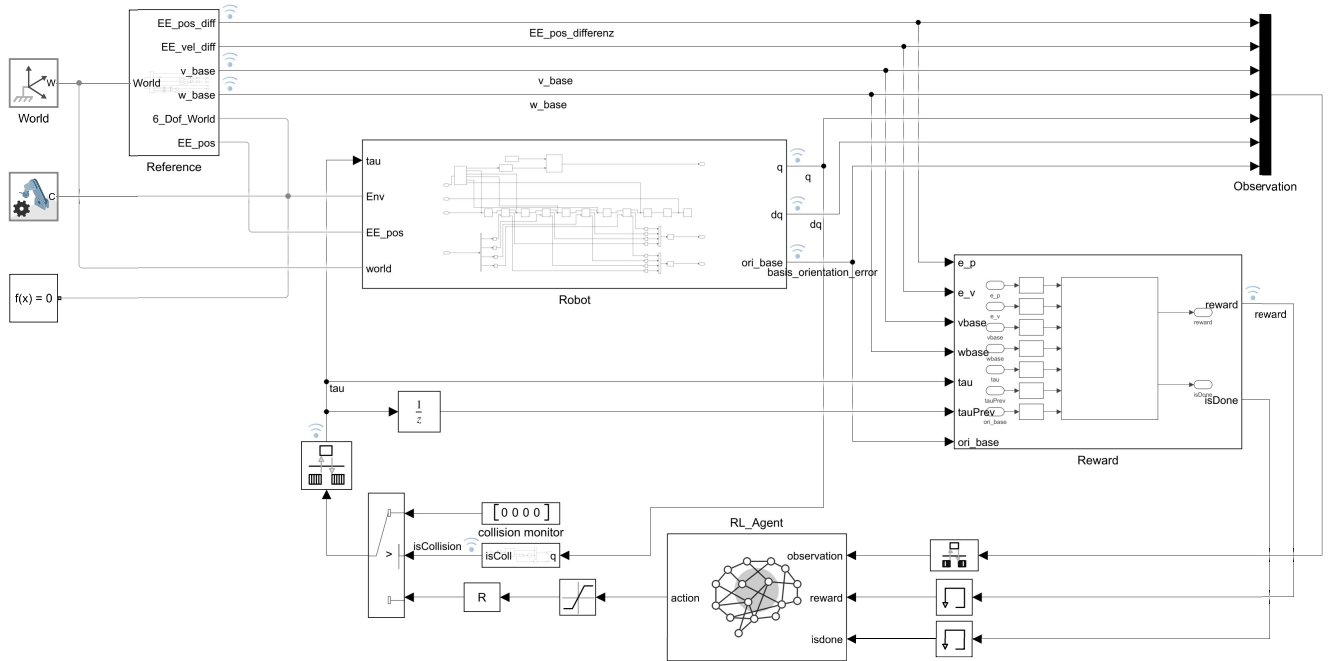


FIGURE 2: Tracking the desired end-effector circular path using inverse kinematics.

FIGURE 3: Overview of the Simulink model `SpaceRobot.slx`. The closed-loop model of the free-floating space robot integrates the robot dynamics, reference generation, observation assembly, reward computation, and a PPO-based RL agent that outputs joint torques. The reward and termination logic evaluate end-effector tracking, base motion, and control effort.

comparison. First, ideal sensing is assumed. All state variables (joint angles, joint velocities, base orientation quaternion, and base angular velocity) are read directly from Simulink signal buses without sensor noise, quantization errors, or measurement bias. Second, zero communication latency is assumed. The RL agent receives observations and issues torque commands within the same discrete time step, with no transmission delay between sensing and actuation. Third, the robot is modeled as a rigid multibody system. Flexible link dynamics, joint compliance, and thermal drift are not included. Fourth, no external disturbances act on the spacecraft. Gravity gradients, atmospheric drag, solar radiation pressure, and residual magnetic torques are all set to zero, consistent with the free-floating assumption in low-

Earth orbit. These assumptions give a controlled benchmark environment. Performance differences can therefore be attributed mainly to the algorithms rather than to environmental variability. Consequently, the reported results characterize relative algorithm performance in a simulation benchmark under controlled assumptions. They do not demonstrate robustness to real sensing, communication latency, structural flexibility, or orbital disturbances, and they do not constitute an assessment of operational readiness for on-orbit deployment. A limited stress test under non-ideal conditions is reported in Section VII-D. The impact of the assumptions on real-world deployment is discussed in Section VIII.

MATLAB/Simulink was chosen as the simulation environment after considering two alternatives that are common in

TABLE 3: Comparison of simulation platforms relevant to free-floating space robotics with formal safety monitors.

Criterion	MATLAB/ Simulink	Gazebo	MuJoCo
Physics engine	Simscape Multibody	ODE, Bullet, DART	MuJoCo (na- tive)
Free-floating multibody fidelity	High	Medium	High
Formal verification integration	Native (Stateflow, Design Verifier)	None	None
RL framework	RL Toolbox	gym-gazebo, ROS bridges	Gymnasium, dm_control
Sensor and ROS inte- gration	Limited (via add-ons)	Extensive	Limited
Code generation / HIL path	Mature (Embedded Coder)	Via ROS 2	Limited
Licensing	Commercial	Open source	Open source

robotics and RL research, Gazebo [50] and MuJoCo [49]. MuJoCo offers an efficient contact and multibody solver and is widely used for RL benchmarks in continuous control [51]. However, it has no native interface to Simulink, Stateflow, and the model-based verification tools that this work uses for the safety components. Gazebo provides mature Robot Operating System (ROS) integration and plug-in-based sensor models and is well suited for hardware-in-the-loop (HIL) development [52]. Its general-purpose multibody back-ends are, however, less convenient for free-floating spacecraft scenarios, and it offers no native path to formal verification tools. Simscape Multibody provides a multibody solver for the free-floating base-manipulator coupling. It is directly connected to Stateflow and Simulink Design Verifier, which this work uses for the safety components. Together with the Reinforcement Learning Toolbox and a deterministic fixed-step solver, this gives a single reproducible toolchain from the physical model to the learned policy and the safety checks. Table 3 summarizes the trade-off. This choice gives up the higher simulation speed of MuJoCo and the richer sensor and ROS tooling of Gazebo in exchange for a close coupling of the dynamics, the RL agent, and the formal-methods tools.

The Simulink model is composed of several subsystems, as illustrated in Fig. 3. The Robot Plant subsystem computes the multibody dynamics of the free-floating robot at each time step (see Fig. 4) by numerically integrating the equations of motion for the applied joint torques [2]. The Controller subsystem executes the learned RL policy. The commanded joint torques pass through a saturation block and a switch before reaching the plant. When the collision monitor reports a contact, the switch overrides the torques to zero [24], [25]. The Reference Trajectory Generator provides the desired end-effector trajectory and base attitude (see Fig. 5), outputting the current target position and velocity at each time step. A set of sensors measures the robot state (joint angles, joint velocities, base orientation, and base angular velocity) and assembles the observation signal for the RL

agent (detailed in Section IV). Finally, two safety monitors check the operational constraints at every agent step. The collision monitor computes the minimum distance between non-adjacent robot bodies and detects contacts. The joint-limit monitor compares every joint angle with its admissible range. A detected contact or a joint-limit violation terminates the episode [24], [25], [39]. Details are given in Section V.

To couple the continuous dynamics with the discrete-time RL agent, a consistent integration scheme is used. The model is simulated with a fixed-step solver (ode4) and a step size of $T_s = 0.01$ s instead of the default variable-step solver. The agent acts every $\Delta t = 0.1$ s. Rate transition blocks connect the two rates and hold each torque command constant over the ten solver steps of an agent interval [40], [41]. Simulink unit consistency checks are enabled to detect modeling errors such as mismatched units early in development.

D. REFERENCE TRAJECTORY AND DESIRED TASK

The task is to follow a semicircular end-effector trajectory while minimizing base disturbance. The reference starts at the end-effector position of the stretched arm at (4.5, 0, 0) m and follows a semicircle with radius $r = 0.5$ m around (4.0, 0, 0) m within $T = 8.5$ s (see Fig. 2). Before training, the trajectory was checked with inverse kinematics for geometric feasibility, that is, for compliance with the joint limits and the absence of singularities. The inverse-kinematics solution gave a consistent set of joint trajectories that realize the desired end-effector motion. The path is therefore kinematically feasible. A failure of the RL agent can thus be attributed to learning and not to a kinematically infeasible reference. During simulation, the reference subsystem outputs the desired end-effector position and velocity as time series $EE_{\text{ref}}(t)$ and $EE_{v,\text{ref}}(t)$ at each time step. The instantaneous position and velocity tracking errors are derived from these signals. They enter both the observation vector and the reward computation that penalizes large errors.

IV. REINFORCEMENT LEARNING FRAMEWORK

State (Observation) Space. At each time step the RL agent receives an observation that summarizes the state variables relevant for control [22]. The state vector consists of the following components:

- *End-effector tracking error.* The Cartesian position error $e_p \in \mathbb{R}^3$ between the current and reference end-effector positions, together with the velocity error $e_v \in \mathbb{R}^3$. When the end-effector is on the reference trajectory, $e_p = 0$. The pair (e_p, e_v) indicates both the spatial deviation from the reference and the velocity mismatch.
- *Arm joint states.* The joint angles $q \in \mathbb{R}^4$ and joint angular velocities $\dot{q} \in \mathbb{R}^4$. These describe the configuration and motion state of the manipulator. Joint angles are provided in radians together with the corresponding mechanical limits.
- *Base motion.* The orientation, angular velocity, and linear velocity of the free-floating base. The base attitude error is computed from the measured base quater-

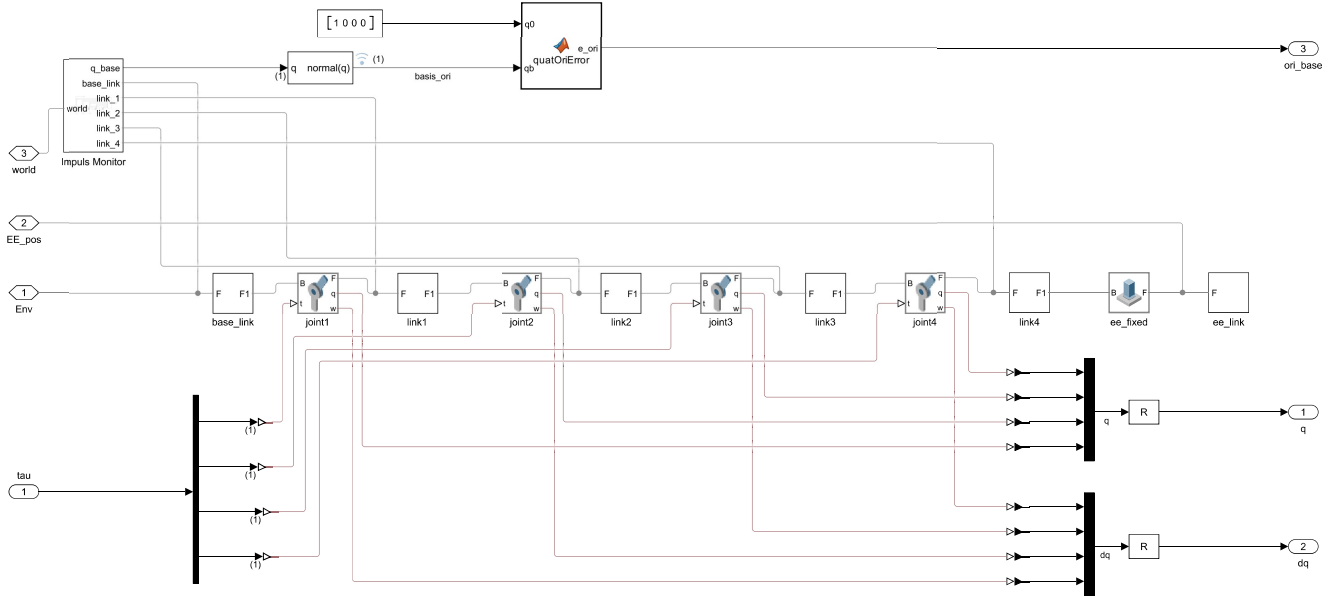


FIGURE 4: Robot subsystem of the Simulink model *SpaceRobot.slx*. The Simscape Multibody model of the free-floating space robot consists of a floating base and a serial 4-DOF manipulator. Joint torque commands are applied to the revolute joints, and the joint positions and velocities are measured and fed back as state variables. The base orientation is represented as a unit quaternion and compared with a reference to compute the base attitude error.

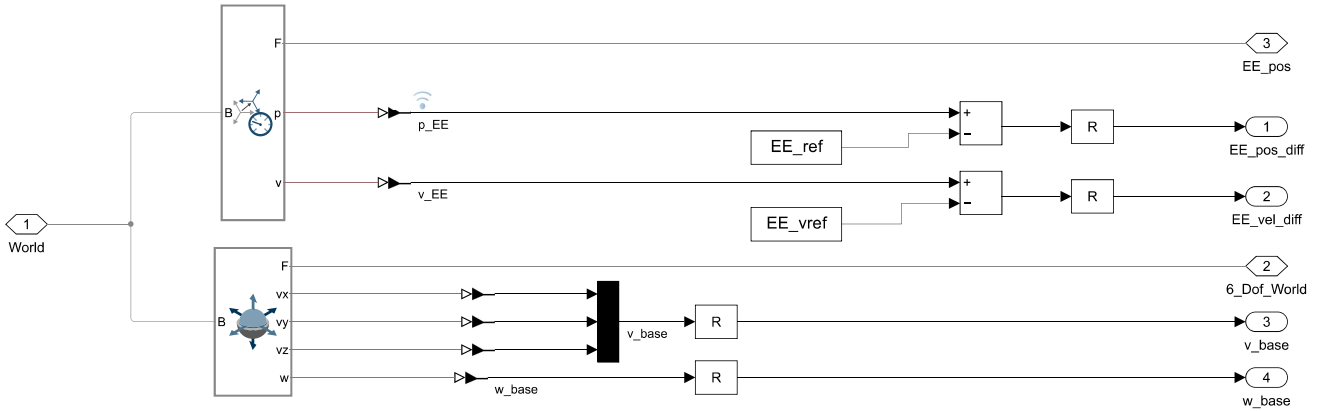


FIGURE 5: Reference subsystem of the Simulink model *SpaceRobot.slx*. The measured end-effector position and velocity are compared with the reference signals. The resulting error vectors are used for the observation and the reward. The base linear and angular velocities are also extracted to capture the base motion induced by the manipulator.

nion q_b and the reference quaternion q_0 as the relative quaternion $q_{err} = q_0^* \otimes q_b$. Its sign is chosen so that the scalar part is non-negative. The relative quaternion is then mapped to the axis-angle rotation vector $e_{ori} = \theta \mathbf{u} \in \mathbb{R}^3$ with $\theta = 2 \operatorname{atan2}(\|q_{err,v}\|, q_{err,w})$ and $\mathbf{u} = q_{err,v} / \|q_{err,v}\|$. This 3-dimensional minimal representation removes the redundancy of the unit-quaternion parameterization and is well-defined for $q_{err} \neq \pm[1, 0, 0, 0]^T$. The base should keep its initial attitude, so the agent has to keep e_{ori} close to zero. The base angular velocity $\omega_{base} \in \mathbb{R}^3$ and the base linear velocity $v_{base} \in \mathbb{R}^3$ are also included to capture the residual rotation and the momentum-coupled translation

of the base.

The overall observation vector has dimension $3(e_p) + 3(e_v) + 4(q) + 4(\dot{q}) + 3(\operatorname{ori}_{base}) + 3(\omega_{base}) + 3(v_{base}) = 23$. This state representation contains the variables required by the agent to assess tracking deviation, arm configuration, and base motion.

Action Space. The action is a vector of continuous joint torque commands. At each decision step, the agent outputs $a = [\tau_1, \tau_2, \tau_3, \tau_4]^T$, with one component per revolute joint of the arm [22], [30]. The actions are real-valued and bounded by the actuator capabilities. Symmetric torque limits $|\tau_i| \leq \tau_{max}$ for $i = 1, \dots, 4$ are enforced through saturation blocks that clip the commanded torque to the interval

$[-\tau_{\max}, +\tau_{\max}]$. The value of $\tau_{\max}$ is selected according to the robot design ($\tau_{\max} = 2 \text{ N m}$ in this work), provided it is large enough to follow the reference trajectory. The saturation enforces both the actuator limits and an upper bound on the applied torque for safety reasons. The action space is therefore $\mathbb{R}^4$, which is compatible with policy gradient and actor-critic methods.

The policy is a mapping from the 23-dimensional state space to the 4-dimensional action space, $\pi : s \mapsto a$, parameterized by a neural network whose parameters are adjusted during training to maximize the expected cumulative reward. The agent issues a new torque command at a fixed interval $\Delta t = 0.1 \text{ s}$. With an episode duration of 8.5 s , this gives $N = 85$ decisions per episode.

Reward Function Design. The reward signal guides the agent toward accurate trajectory tracking. At the same time, it penalizes base disturbance and favors smooth and bounded control actions. At each discrete time step t , the reward r_t is computed as the sum of a progress term and a proximity bonus, minus a weighted cost function:

$$r_t = \frac{1}{C} (r_{\text{prog}} + r_{\text{bonus}} - c_t), \quad (1)$$

where C is a normalization constant.

To encourage monotonic convergence toward the reference trajectory, a progress reward is defined based on the reduction of the end-effector position error:

$$r_{\text{prog}} = \begin{cases} k_p (d_{t-1} - d_t), & d_{t-1} > d_t, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where $d_t = \|e_p(t)\|$ is the Euclidean end-effector position error and k_p is a scaling factor. The previous distance d_{t-1} is stored persistently across time steps.

A small shaping bonus is added when the end-effector is close to the reference point:

$$r_{\text{bonus}} = k_b \exp\left(-\left(\frac{d_t}{\sigma}\right)^2\right), \quad (3)$$

which is intended to improve the local convergence near the desired trajectory.

The instantaneous cost c_t penalizes tracking errors, base motion, and control effort:

$$c_t = W_p \|e_p\|^2 + W_v \|e_v\|^2 + W_{\text{ori}} \|e_{\text{ori}}\|^2 + W_{\omega_b} \|\omega_{\text{base}}\|^2 + W_{v_b} \|v_{\text{base}}\|^2 + W_u \|\tau\|^2 + W_d \|\tau - \tau_{t-1}\|^2, \quad (4)$$

where e_p and e_v denote the end-effector position and velocity errors, e_{ori} the base orientation error, v_{base} and ω_{base} the linear and angular base velocities, and τ the joint torque vector. The final term penalizes changes in control input to promote smooth actuation.

The weights used in all experiments are summarized in Table 4. The physical intuition behind the weight choices is as follows. The end-effector position error weight $W_p = 150$ is large because trajectory tracking is the primary task objective. A high weight makes the position error the dominant

term of the cost at typical error magnitudes. The velocity error weight $W_v = 25$ is lower than W_p to avoid over-penalizing transient dynamics during fast arm motion, while still discouraging persistent velocity offsets. The base orientation weight $W_{\text{ori}} = 200$ is the largest weight in the cost function, because base attitude disturbance is safety-critical in on-orbit operations. An uncontrolled rotation of the spacecraft base can endanger the mission regardless of the tracking performance. The base angular velocity weight $W_{\omega_b} = 8$ penalizes ongoing base rotation, similar to a damping term. The linear velocity weight $W_{v_b} = 2$ is kept small. With zero initial linear momentum, base translation is coupled to the manipulator motion and is expected to remain small for the considered task. It cannot be avoided completely, because the center of mass of the whole system stays fixed. The torque magnitude weight $W_u = 0.02$ and the torque-rate weight $W_d = 0.06$ are small so that the agent keeps enough freedom to explore different control actions. The torque-rate penalty is slightly larger to discourage high-frequency chattering in the control signal. The progress scaling factor $k_p = 10$ is chosen so that the progress reward has a magnitude comparable to the cost terms at typical error levels and is not dominated by them. The proximity bonus parameters $k_b = 0.05$ and $\sigma = 0.02$ are small by design. They provide a weak local guidance signal near the trajectory without creating a strong local attractor that could trap the policy. Finally, the normalization constant $C = 500$ scales the total reward to a range of approximately $[-1, 0]$ under typical operating conditions, which is intended to stabilize the value-function learning of all tested algorithms.

At each step, all inputs to the reward function are checked for numerical validity. An episode is terminated early if a NaN or Inf value is detected, if the end-effector error exceeds 10 m , if the collision monitor reports a contact, or if a joint leaves its admissible range. In that case, the agent receives the lower end of the one-step reward range, -1 , for the terminal step and for every remaining step of the episode. The terminal penalty is therefore

$$r_{\text{fail}} = -(N - k + 1), \quad (5)$$

where k is the step at which the violation occurs and $N = 85$ is the number of steps per episode. A constant penalty is not sufficient in this setting. The step rewards are negative in nearly all states, so a single penalty of -1 would make an early termination more attractive than continuing with a poor policy. In a pilot run with a constant penalty of -1 , none of the last 100 training episodes of the default and the optimized PPO agent reached the end of the horizon. With the penalty in (5), all of them did. All reward components were implemented in a MATLAB Function block within the Simulink environment.

Training Procedure. An episodic training scheme is used. Each episode covers the semicircular reference once and lasts $T = 8.5 \text{ s}$, which corresponds to $N = 85$ agent steps. At the start of each episode, a custom reset function sets the base to its home orientation at rest. The initial joint angles are

drawn uniformly within $\pm 1^\circ$ around the nominal stretched configuration $q = 0$, in which the end-effector is located at the start of the reference. These initial states lie close to a configuration that is free of collisions and far from the joint limits. All state variables, integrator states, and logging signals are reset as well [22].

Within an episode, the agent receives an observation at each step and outputs a torque action. The simulation runs until the end of the horizon or until one of the terminal conditions of the reward function occurs. In that case the penalty in (5) is applied and the episode ends at that step.

The training loop is provided by MATLAB's Reinforcement Learning Toolbox. After each episode or after a fixed number of steps, the collected states, actions, and rewards are used to update the policy according to the chosen RL algorithm [42]. Where applicable, the value function is updated as well. The algorithm-specific update rules of PPO, DDPG, and the remaining methods are handled by their respective toolbox implementations. This work supplies the experience and the reward signal [22]. All agents are trained for the same number of episodes, $N_{\text{train}} = 1000$. This budget is intended to let most algorithms converge or reach a performance plateau. During training, the cumulative reward and the length of every episode are recorded. A successful training run is characterized by an episode return that increases toward zero, since the step rewards are negative costs. An episode that ends before N steps was terminated by a monitor, so the episode length also shows how often safety violations occur during training.

To make the runs reproducible, MATLAB's Mersenne Twister random number generator is initialized with a seed s before each training run (`rng(s, 'twister')`). The seed determines the initial network weights, the exploration noise, the sampling of mini-batches, and the initial joint angles of every episode. Two runs with the same seed are identical. Each configuration was trained with ten seeds ($s = 0, \dots, 9$). This results in 70 training runs for the six default agents and the optimized PPO agent. Ten seeds are used so that the seed-level tests keep some resolution after the correction for multiple comparisons. Every run comprises exactly $N_{\text{train}} = 1000$ episodes without early stopping, and the policy after the last episode is evaluated. No checkpoint is selected, and all ten seeds are reported.

TABLE 4: Reward function weights used in all experiments.

Term	Symbol	Value
End-effector position error	W_p	150
End-effector velocity error	W_v	25
Base orientation error	W_{ori}	200
Base angular velocity	W_{ω_b}	8
Base linear velocity	W_{v_b}	2
Joint torque magnitude	W_u	0.02
Torque rate (smoothness)	W_d	0.06
Progress scaling factor	k_p	10
Proximity bonus scaling	k_b	0.05
Proximity width	σ	0.02
Reward normalization constant	C	500

Evaluation and Metrics. After training, each policy is evaluated deterministically. For the stochastic policies, the mean action is used. Every policy is evaluated on $N_{\text{eval}} = 30$ episodes whose initial joint angles are drawn uniformly within $\pm 1^\circ$ from a fixed random stream, and on one additional episode from the nominal start pose. All agents use the same initial states, so the evaluation episodes are paired across agents. The conclusions therefore remain conditional on these initial states and the tested reference trajectories. Over the evaluation episodes, a set of Key Performance Indicators (KPIs) is computed to compare the agents quantitatively. For episodes that terminate early due to a safety violation, each KPI is computed over the executed portion of the episode up to the termination step. The KPIs are first computed for each seed over the 30 random episodes. The seed dependence and the performance reached are then reported separately. A training run counts as successful when none of its 31 evaluation episodes is terminated by a monitor. The first table reports the number of successful runs per configuration and the share of evaluation episodes that a monitor terminated, taken over all runs. The KPI tables report the mean and the standard deviation over the successful runs only, together with their number. These KPI values are therefore conditional on a successful run. Algorithms are compared on the seed level with two-sided Mann–Whitney U tests over the successful runs and a Holm correction for multiple comparisons. A test is reported only when both configurations have at least four successful runs. The shares of successful runs are compared with Fisher's exact test, also with a Holm correction.

The nine KPIs are organized into three groups to aid interpretation. The tracking accuracy group (K_2 and K_3) measures how closely the end-effector follows the reference path. The base stability and safety group covers spacecraft attitude drift (K_4) and constraint violations (K_5 and K_6). The efficiency group covers control smoothness (K_7), residual base rotation (K_8), and energy consumption (K_9). K_1 is an overall scalar summary of the agent's performance across all objectives, since it is the cumulative reward that jointly penalizes all undesired behaviors.

- **K_1 (Mean Episodic Return).** Mean cumulative reward per episode. Higher is better and reflects overall task success.

$$R_i = \sum_{k=1}^{T_i} r_i[k], \quad K_1 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} R_i.$$

- **K_2 (Mean Squared EE Position Error).** Average squared tracking error per episode. Lower is better.

$$K_2 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{p,i}[k]\|_2^2.$$

- **K_3 (Maximum EE Position Error).** Worst-case tracking deviation. Lower is better.

$$K_3 = \max_i \max_k \|e_{p,i}[k]\|_2.$$

- **K_4** (Mean Base Orientation Error). Average attitude drift of the spacecraft base, measured as the magnitude of the axis-angle rotation-vector error e_{ori} (in radians). Lower is better.

$$K_4 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{\text{ori},i}[k]\|_2.$$

- **K_5** (Collision Rate). Fraction of episodes with at least one collision. Zero is ideal.

$$C_i = \mathbb{I}(\exists k : c_i[k] > 0.5), \quad K_5 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} C_i.$$

- **K_6** (Joint Limit Violation Rate). Fraction of time steps across episodes where any joint exceeds its mechanical limit. Zero is ideal.

$$K_6 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i n_J} \sum_{k=1}^{T_i} \sum_{j=1}^{n_J} \mathbb{I}(q_{i,j}[k] \notin [q_j^{\min}, q_j^{\max}]).$$

- **K_7** (Control Smoothness (Jerk)). Second difference of joint torques. Lower values indicate smoother, less chattering actuation.

$$\Delta^2 \tau_i[k] = \tau_i[k+2] - 2\tau_i[k+1] + \tau_i[k],$$

$$J_i = \frac{1}{T_i - 2} \sum_{k=1}^{T_i-2} \|\Delta^2 \tau_i[k]\|_2,$$

$$K_7 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} J_i.$$

- **K_8** (Mean Base Angular Velocity). Residual spacecraft rotation rate during the task (rad/s). Lower is better, ideally zero.

$$K_8 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|\omega_{b,i}[k]\|_2.$$

- **K_9** (Energy Consumption). Average absolute mechanical power integrated over each episode. Lower is better when not at the expense of tracking quality.

$$E_i = \int \sum_{j=1}^{n_J} |\tau_{i,j}(t) \dot{q}_{i,j}(t)| dt, \quad K_9 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} E_i.$$

Two KPIs need a justification in the context of space missions. The control smoothness K_7 captures high-frequency torque oscillations (chattering). In space, such oscillations can accelerate actuator wear and excite structural vibrations. These vibrations degrade the pointing accuracy of sensitive instruments such as cameras or star trackers. Rapid back-and-forth motion also dissipates energy. Minimizing K_7 therefore supports hardware lifetime and payload pointing requirements. The energy consumption K_9 reflects the mechanical work performed by the joints, approximated as $\int \sum_j |\tau_j \dot{q}_j| dt$. Onboard power is limited by solar panel area and battery capacity. Energy spent on control actuation reduces the budget available for communication, thermal

management, and payload operation. Reducing K_9 without sacrificing tracking accuracy therefore has operational value for mission planning.

In addition to the KPIs above, two training-related metrics are reported. T_1 is the standard deviation of the episode reward over the last 100 training episodes, an indicator of training stability. T_2 is the wall-clock training time required for 1000 episodes, an indicator of the computational load.

Applying these metrics to all six agents yields a common basis for comparison. The comparative results are presented and discussed in Section VI.

V. SAFETY MONITORING AND CONSTRAINT ENFORCEMENT

Stability and safety are central requirements in space robotics. A controller for autonomous on-orbit servicing must satisfy the physical constraints of the system and avoid hazardous states, in particular when the control policy is learned from data and must therefore handle uncertainty. In this work, runtime safety monitors and constraint-enforcement mechanisms are integrated directly into the RL training and execution environment. They have two purposes. They terminate episodes that reach unsafe states during exploration, and they shape the agent's behavior toward the monitored constraints through penalties [22], [24]. These mechanisms address a defined set of properties and do not constitute a formal safety certification of the learned controller (see the limitations at the end of this section).

Momentum Conservation Monitor. In a free-floating system without external forces, both the total linear and the total angular momentum of the robot (base and arm) are conserved. With zero initial momentum, the total linear momentum remains close to zero throughout the simulation [2]. A momentum observer computes the total linear momentum p_{tot} and checks for its conservation (see Fig. 6). Virtual sensors attached to the base and to each link provide the body velocities, from which $p_{\text{tot}} = \sum_i m_i v_i$ is computed. In an ideal simulation, p_{tot} remains close to zero throughout an episode. Small numerical deviations may arise from time-integration errors. A persistent, systematic drift of p_{tot} would indicate an unintended external force or a modeling error, such as an unmodeled constraint. Fig. 7 shows p_{tot} for an evaluation episode of the optimized PPO agent from the nominal initial state. An offset of 2.7×10^{-7} N s appears in the first agent step, after which p_{tot} stays constant. For TRPO and the optimized PPO agent, $|p_{\text{tot}}|$ remained below 10^{-6} N s in all 620 evaluation episodes. The other agents reached up to 4.3×10^{-4} N s. These larger deviations may be explained by the integration error of the fixed-step solver for rapidly changing torque commands. The values indicate that the simulation conserves momentum within the numerical tolerance. The momentum observer is a diagnostic tool. It is evaluated offline and does not influence the training.

Collision Monitor. At every agent step, a collision monitor computes the minimum distance d_{min} between non-adjacent robot bodies from the URDF collision geometry

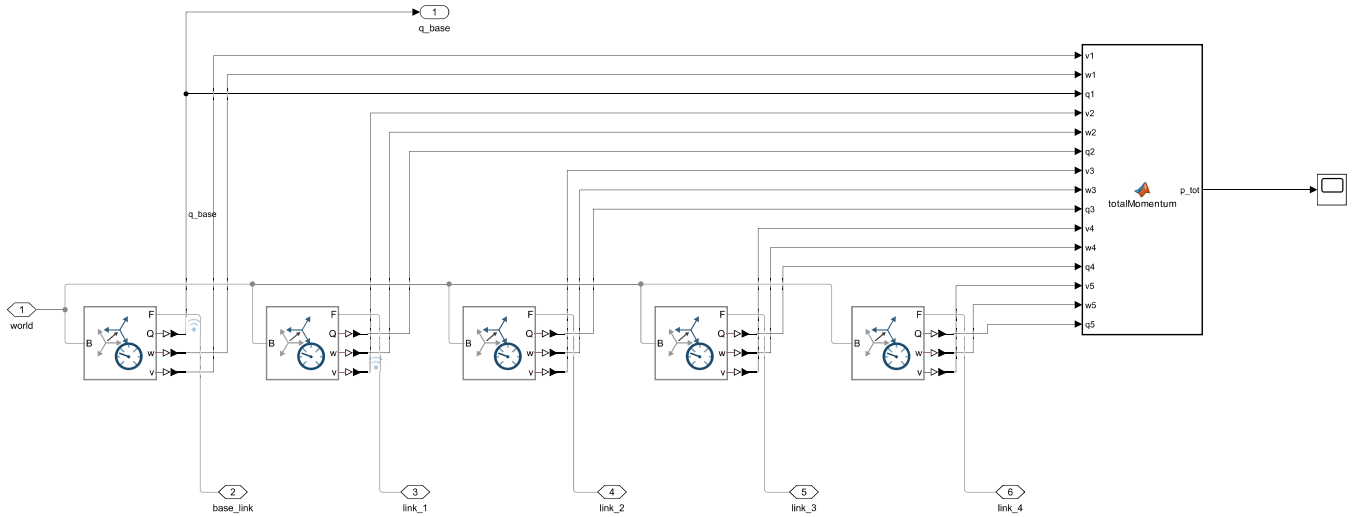


FIGURE 6: Momentum monitor subsystem (momentum observer) of the Simulink model `SpaceRobot.slx`. The velocities and orientations of the base and of all manipulator links are measured and combined in a MATLAB function to compute the total linear momentum of the free-floating space robot. The monitor is used to check momentum conservation in the simulation.

(see Fig. 8) [39]. If $d_{\min}$ falls below the safety threshold $d_{\text{safe}} = 2$ cm, a warning is raised and logged. If a contact between two bodies is detected, all commanded joint torques are overridden to zero and the episode is terminated with the penalty in (5). The monitor was tested by intentionally commanding the arm toward the base and by starting an episode in a colliding configuration. In both cases, the contact was detected, and the robot was stopped (see Fig. 9). During training, collisions occurred mainly in early exploratory episodes. The monitor detected them and terminated the episode. In the evaluation of the 70 trained agents, 108 of 2170 episodes ended with a detected contact. These episodes belong to six runs of DDPG, TD3, PG, and the default PPO agent. No contact occurred in the evaluation episodes of TRPO, SAC, and the optimized PPO agent.

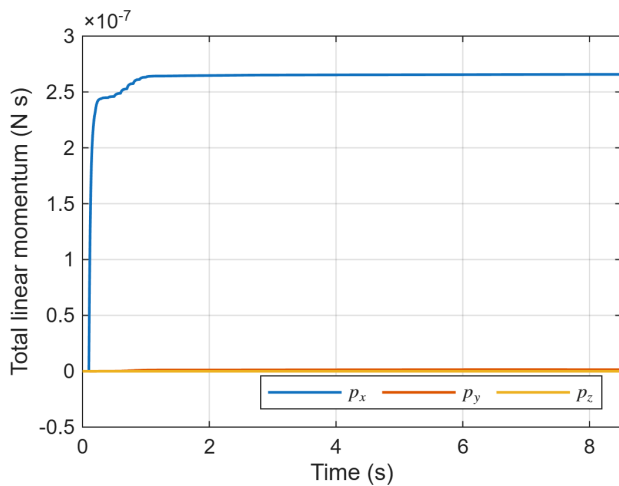


FIGURE 7: Total linear momentum of the robot during an episode.

Joint Limit Enforcement. Each revolute joint of the arm is subject to mechanical limits. Violating these limits could damage a real robot and often indicates an unstable controller. The joint limits are enforced at two levels. First, the Simscape model includes mechanical joint limits with penalty forces, i.e., a stiff barrier that opposes further motion beyond the limit. Second, a joint-limit monitor in the reward block compares every joint angle with its admissible range at each agent step. A violation terminates the episode with the penalty in (5), which prevents further motion into the limit. In the evaluation of the 70 trained agents, joint-limit violations terminated 791 of the 2170 episodes. They were the most frequent reason for a terminated episode. TRPO and the optimized PPO agent reached no joint limit in any evaluation episode. Because a violation ends the episode, the joint limit violation rate K_6 is zero for every run without a terminated episode. The violations are therefore reported as abort rates in Table 7.

Formal Verification of Torque Constraints. In addition to runtime saturation, an offline formal verification was performed to check torque-bound compliance of the trained policy block. Using MATLAB Simulink Design Verifier (SLDV) in property-proving mode, a simplified closed-loop model of the agent and the joint torques was analyzed. The property to be checked is that, for any admissible observation input in the analyzed abstraction, each commanded torque τ_i satisfies $\tau_i \in [-\tau_{\max}, +\tau_{\max}]$ (see Fig. 10). SLDV found no counterexample for this property. The corresponding SLDV result is shown in Fig. 11 [23]. This result applies to the simplified verification model, including admissible abstract states not visited during simulation. The full Simscape model is not amenable to property proving because of the nonlinear dynamics. The verification was performed on a simplified model (`Verify_tau.slx`) in which the plant was re-

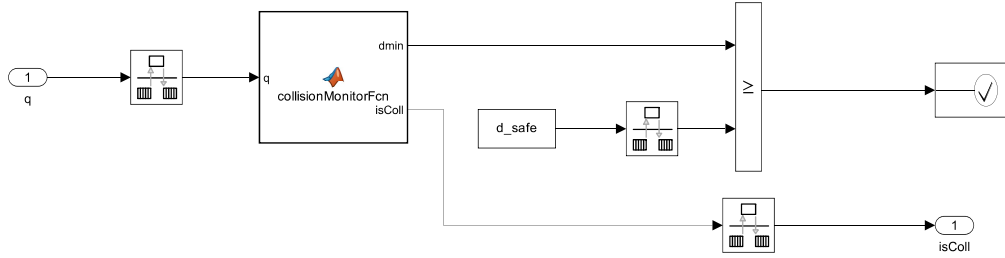


FIGURE 8: Collision monitor subsystem of the Simulink model `SpaceRobot.slx`. The joint and link configurations are used to compute the minimum distance between non-adjacent robot bodies. A distance below the safety threshold raises a warning. A detected contact sets the collision flag, which terminates the episode.

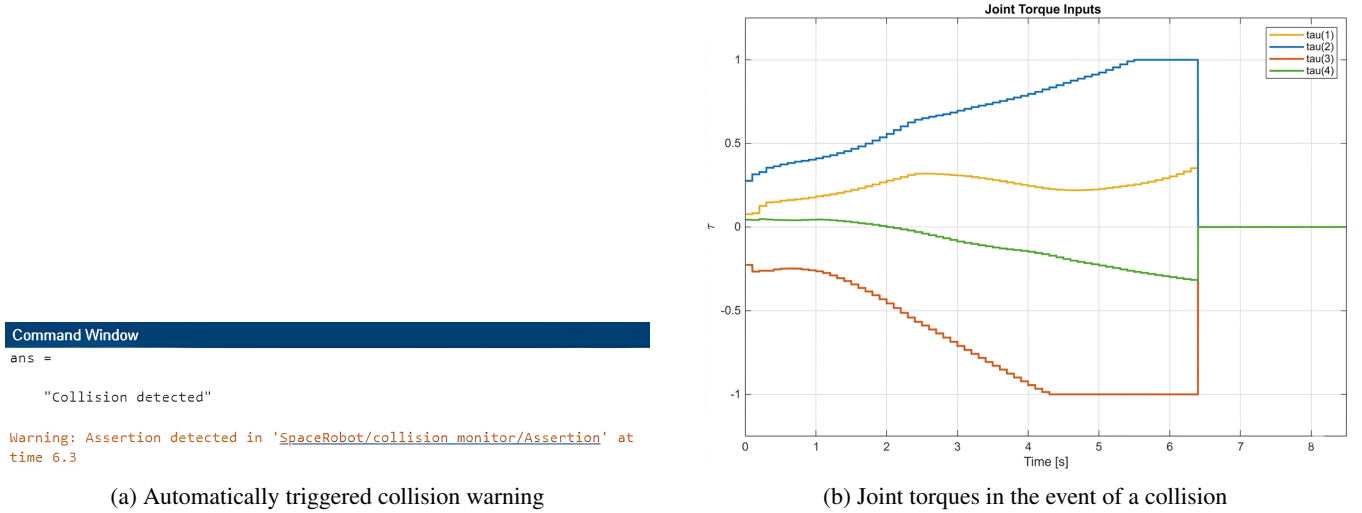


FIGURE 9: Test of the collision monitor and the associated joint torques.

placed by a worst-case abstraction, because only the policy output block is relevant for the considered property. This illustrates how formal methods can complement simulation-based learning by checking selected properties on an abstracted model of the agent.

Limitations of the Formal Methods Component. The mechanisms above cover a defined set of monitored properties, namely collision events, joint-limit events, bounded torques, and physical consistency through the momentum check. Other properties, such as task completion within a time bound or formal stability margins, are not verified. In particular, the runtime monitors detect violations and terminate the episode but do not prove their absence, and the offline SLDV check establishes only the torque bound on the simplified verification model. Neither closed-loop stability, task completion, nor collision avoidance over all possible trajectories of the full nonlinear Simscape model is formally verified. A complete formal specification of the mission, for example temporal-logic specifications such as “the robot shall accomplish X without Y happening”, would be considerably more complex and is beyond the scope of this work. This work targets the safety properties that are considered most relevant for the task. Within this scope, the runtime monitors terminated episodes that would have

violated a monitored property and penalized the corresponding actions. For TRPO and the optimized PPO agent, no violations of the monitored constraints were observed in any evaluation episode [25]. The other agents violated a monitored constraint in part of their runs, and the monitors terminated these episodes. This is empirical evidence for the tested initial states and reference trajectories, not a guarantee over all reachable states.

Computational Overhead and Scalability. The momentum observer and the joint-limit monitor consist of simple algebraic operations over the link states and scale as $\mathcal{O}(n_{\text{links}})$ per time step. The pairwise distance check of the collision monitor scales as $\mathcal{O}(n_{\text{links}}^2)$. For the 4-DOF arm considered here, this overhead is small compared with the physics simulation itself. End-to-end training times, including the full safety stack, are reported as metric T_2 in Table 8. They range from about 7 minutes (PG) to 35 minutes (SAC) for 1000 episodes on a standard workstation (AMD Ryzen 7 7700, 32 GB RAM), with six training runs executed in parallel. For onboard deployment, only the learned policy network needs to execute in the control loop. The default and the optimized PPO agents use a compact policy network with two hidden layers of 128 units and about 2.1×10^4 parameters. This is well below the memory footprint of typical perception

networks deployed on space-qualified hardware. At the 10 Hz control rate used here, a single forward pass through this network requires about 2×10^4 multiply-accumulate operations per step. This load appears compatible with the throughput of modern space-qualified embedded processors, for example ARM Cortex-A or LEON-class CPUs with a hardware floating-point unit. The safety monitors can be retained as lightweight parallel watchdog processes or reduced to the most safety-critical checks (torque saturation and joint-limit monitor) with little additional computation. This modular structure may simplify the transfer to resource-constrained onboard systems.

In summary, the runtime safety monitors integrated into the learning process terminated unsafe simulated states during the exploratory phase of training and shaped the agent's behavior by penalizing unsafe actions. The agent therefore learns under runtime monitoring of the selected constraints. This combination of monitored runtime constraints and penalty-driven learning is one approach to training deep RL policies in safety-critical domains such as space robotics, in which unconstrained trial-and-error exploration is not admissible. The next section analyzes the performance of the six RL agents under these conditions.

VI. COMPARISON OF CONTINUOUS RL AGENTS

This section compares six continuous-action RL algorithms from three families on the same task and environment [22], [30]. The following algorithms are considered:

- *Policy Gradient (PG)*. The REINFORCE algorithm (Monte Carlo policy gradient), included as a baseline for

direct policy optimization without variance-reduction techniques [21], [22].

- *Proximal Policy Optimization (PPO)*. A policy-gradient method that uses a clipped probability-ratio surrogate to bound policy updates [20].
- *Trust Region Policy Optimization (TRPO)*. A policy-gradient method that enforces a trust-region constraint based on the Kullback–Leibler divergence at each update. The constrained optimization step incurs a higher per-update computational cost than PPO [17].
- *Deep Deterministic Policy Gradient (DDPG)*. An off-policy actor-critic algorithm with a deterministic policy and target networks. DDPG jointly learns a Q -function and a policy and is sensitive to hyperparameter choices [43].
- *Twin Delayed DDPG (TD3)*. A variant of DDPG that uses two critic networks to reduce overestimation bias and delays policy updates relative to value updates [18].
- *Soft Actor-Critic (SAC)*. An off-policy actor-critic algorithm with a stochastic policy and an entropy term in the objective. This entropy regularization promotes exploration but increases the per-step computational cost relative to deterministic methods [19].

The six algorithms cover the on-policy/off-policy and deterministic/stochastic axes, as well as a range of update rules. Given the strong base–manipulator coupling and the safety constraints, methods with constrained or clipped policy updates (PPO, TRPO) are expected to provide more stable training than algorithms without such constraints (PG, DDPG). The off-policy actor-critic methods (DDPG, TD3, SAC) may achieve comparable performance but are typically more sensitive to hyperparameter selection.

On-policy methods (PG, PPO, TRPO) update the policy only from data generated by the current policy, which typically leads to stable but less sample-efficient learning. Off-policy methods (DDPG, TD3, SAC) reuse experience stored

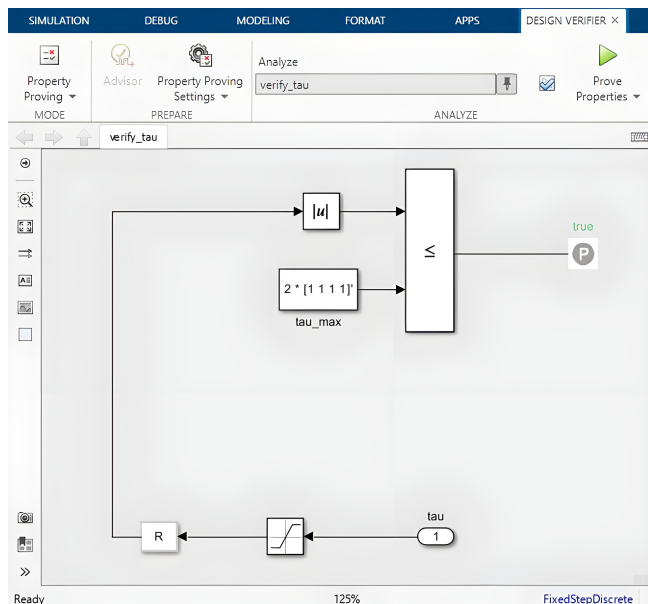


FIGURE 10: Property model for Simulink Design Verifier (Verify_tau.slx). The absolute joint torques are compared with the torque limit using a relational operator. The property is checked over the analyzed abstract state space of the simplified verification model.

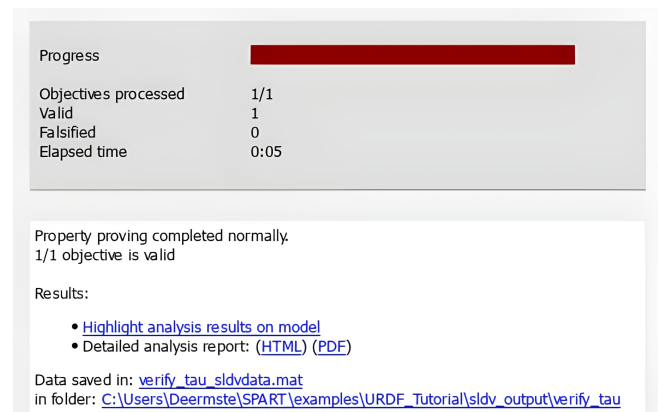


FIGURE 11: Result of the Simulink Design Verifier for the torque-bound property. The analysis found no counterexample for any analyzed execution path of the simplified verification model. The result does not cover other properties or the full nonlinear Simscape model.

TABLE 5: Classification of the evaluated reinforcement learning algorithms.

Algorithm	Policy Type
Policy Gradient (PG)	On-policy
Proximal Policy Optimization (PPO)	On-policy
Trust Region Policy Optimization (TRPO)	On-policy
Deep Deterministic Policy Gradient (DDPG)	Off-policy
Twin Delayed DDPG (TD3)	Off-policy
Soft Actor-Critic (SAC)	Off-policy

in a replay buffer, which improves sample efficiency at the cost of higher sensitivity to hyperparameters [22], [30].

Training Setup. All experiments were performed on a workstation with an AMD Ryzen 7 7700 processor and 32 GB of RAM, using MATLAB R2026a with the Reinforcement Learning Toolbox and Simscape Multibody. All six agents use the default hyperparameters and the default networks of the Reinforcement Learning Toolbox with two hidden layers of 128 ReLU units. Only the sample time is set to Δt . Table 6 lists the resulting values. The reward function, the network size, and the training budget are the same for all agents, and no agent is tuned individually. Observed differences therefore reflect the algorithms together with their default hyperparameters. PPO was further tuned in a subsequent step. In the main comparison, PPO uses its default configuration. Each configuration was trained with ten seeds for 1000 episodes, and the policy after the last episode was evaluated.

Evaluation Protocol. The agents were evaluated with the protocol of Section IV. Each of the 60 default runs was evaluated on the same 31 initial states. A run counts as successful when none of its evaluation episodes was terminated by a monitor. Table 7 reports the number of successful runs and the abort rate over all runs. Table 8 reports the KPIs over the successful runs only.

Success Across Seeds. The number of successful runs ranged from two to ten out of ten (Table 7). TRPO completed all ten runs without a terminated evaluation episode. SAC, TD3, and DDPG succeeded in four to six runs, the default PPO agent in three, and PG in two. In 27 of the 30 failed runs, all 31 evaluation episodes were terminated. A failed run therefore reflects a systematic property of the learned policy. Most terminations were caused by joint-limit violations. Contacts between robot bodies occurred only in runs of TD3, DDPG, PPO, and PG. Compared with TRPO, the number of successful runs was significantly lower for PG, PPO, and DDPG (Fisher's exact test, Holm-corrected $p \leq 0.033$). The differences to TD3 and SAC were not significant with ten runs per agent.

Performance of the Successful Runs. Table 8 summarizes the tracking error K_2 , the base orientation error K_4 , and the training time T_2 of the successful runs. TRPO, PPO, and PG reached mean squared tracking errors between 0.013 and 0.018 in their successful runs. TD3, DDPG, and SAC remained between 0.089 and 0.217, although their successful runs completed every evaluation episode. Their base orien-

tation errors were also larger (K_4 between 0.098 and 0.353, compared with 0.026 for TRPO). The standard deviation of TRPO results mainly from three of the ten runs with K_2 between 0.021 and 0.066 (Fig. 12a). The median of TRPO is 0.0061.

TRPO required 11.3 min per training run, the default PPO agent 8.4 min, and PG 6.9 min. The off-policy agents required 24.3 to 35.2 min. The late-training reward variance was small for TRPO, PPO, and PG ($T_1 \leq 1.23$) and large for TD3, DDPG, and SAC (T_1 between 13.9 and 20.0). This suggests that the off-policy agents had not settled after 1000 episodes.

The control smoothness K_7 and the energy consumption K_9 separate the agents in a similar way. TRPO and PG produced smooth torque commands ($K_7 \leq 0.010$) with a low energy consumption ($K_9 \leq 2.0$). The default PPO agent tracked the path with a similar error but with a larger jerk ($K_7 = 0.383$) and energy consumption ($K_9 = 12.4$). The off-policy agents combined larger tracking errors with K_7 between 0.20 and 0.55 and K_9 between 8.6 and 19.3.

Statistical Tests. Table 9 compares the successful runs of each default agent with those of TRPO on the seed level. PPO and PG are not tested because they have fewer than four successful runs. SAC, TD3, and DDPG had a significantly lower return K_1 and significantly larger tracking errors K_2 and K_3 than TRPO (Holm-corrected $p \leq 0.005$). Their base orientation error K_4 was also significantly larger ($p \leq 0.031$). For SAC, the differences in K_7 and K_9 were not significant.

Fig. 12 shows the distribution of four KPIs over the successful runs, and Fig. 13 shows the corresponding 95% bootstrap confidence intervals of the mean. Both figures include the optimized PPO agent of Section VII for reference.

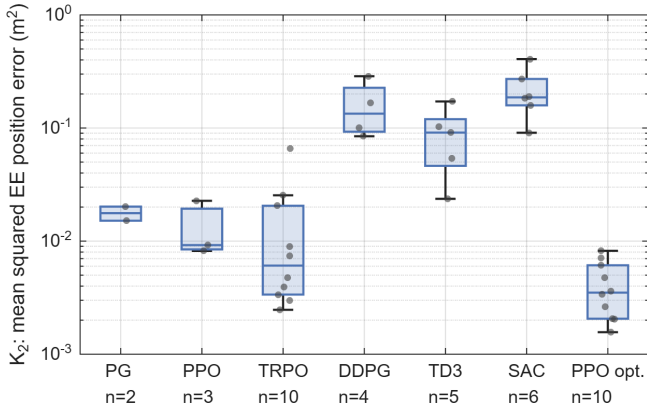
Summary of the Comparison. With the toolbox default hyperparameters, the comparison can be summarized as follows:

- TRPO was the only default agent that completed all ten runs without a terminated evaluation episode. It also reached a low tracking error, although three of its runs were weaker than the others.
- The default PPO agent and PG succeeded in only three and two of ten runs. Their successful runs tracked the path with an error similar to that of TRPO.
- SAC, TD3, and DDPG succeeded in four to six runs. Even in their successful runs, they showed significantly larger tracking errors than TRPO and required two to three times longer training.

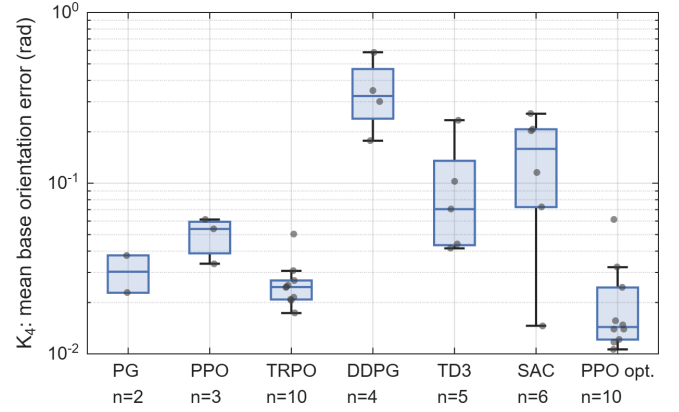
These results suggest that the trust-region constraint of TRPO stabilizes training under the strong base-manipulator coupling. The clipped updates of PPO did not provide the same stability with the default learning rate of 10^{-2} . This learning rate is high for PPO and the off-policy methods and likely contributes to their seed dependence. PPO is selected for tuning in Section VII because its successful runs were competitive with TRPO and it trained faster than TRPO. The tuning therefore targets the seed dependence.

TABLE 6: Hyperparameters of the benchmarked agents. The six default agents use the default values of the Reinforcement Learning Toolbox (R2026a). All agents act every $\Delta t = 0.1$ s and use networks with two hidden layers of 128 ReLU units. α_a and α_c denote the actor and critic learning rates.

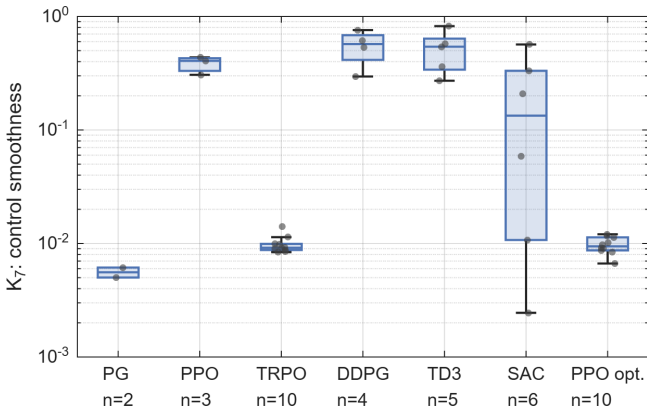
Agent	α_a	α_c	γ	Experience	Mini-batch	Further settings
PG	10^{-2}	10^{-2}	0.99	1 episode	–	entropy weight 0, critic as baseline
PPO (default)	10^{-2}	10^{-2}	0.99	horizon 512	128	3 epochs, clip 0.2, entropy 0.01, GAE 0.95
TRPO	–	10^{-2}	0.99	horizon 512	128	KL limit 0.01, entropy 0.01, GAE 0.95
DDPG	10^{-2}	10^{-2}	0.99	buffer 10^4	64	target smoothing 0.001, OU noise $\sigma = 0.3$
TD3	10^{-2}	10^{-2}	0.99	buffer 10^4	64	target smoothing 0.005, noise $\sigma = 0.32$, policy delay 2
SAC	10^{-2}	10^{-2}	0.99	buffer 10^4	64	target smoothing 0.001
PPO (optimized)	10^{-3}	5×10^{-4}	0.995	horizon 1024	256	10 epochs, clip 0.2, entropy 0.001, GAE 0.95, gradient threshold 1



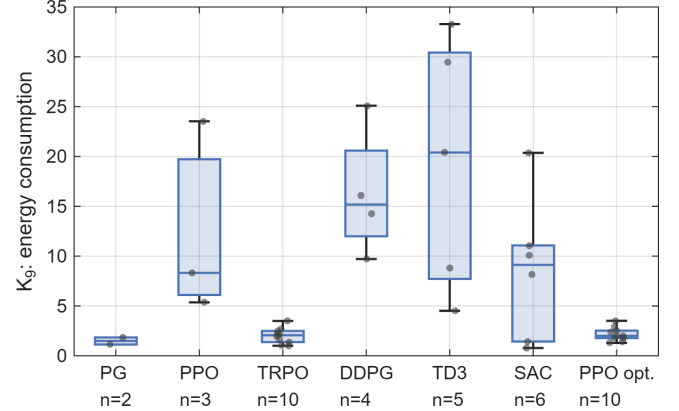
(a) K_2 : Mean Squared EE Position Error



(b) K_4 : Mean Base Orientation Error



(c) K_7 : Control Smoothness (Jerk)



(d) K_9 : Energy Consumption

FIGURE 12: Distribution of selected KPIs over the successful runs of each agent. Each point is one run, averaged over the 30 random initial states, and the number of successful runs is given below each agent. Each box spans the interquartile range, and the center line marks the median. Axes are logarithmic where the values span more than a factor of 50. The optimized PPO agent of Section VII is included for reference.

VII. OPTIMIZATION OF THE PPO AGENT

The default PPO agent succeeded in only three of ten runs in Section VI, but its successful runs tracked the path with an error similar to that of TRPO. This section examines whether a tuned configuration removes this seed dependence. The PPO agent used so far corresponds to the default configuration of MATLAB's Reinforcement Learning Toolbox [42]. The tuning concerns the hyperparameters, while the hidden layers of the networks are kept.

Network Architecture. The optimized agent keeps the hidden layers of the default PPO agent. The policy and the value network both use two hidden layers of 128 units with ReLU activations. In contrast to the default agent, the mean output of the Gaussian policy is not bounded by a tanh layer. The commanded torques are limited by the saturation block instead. The networks are defined explicitly rather than through the toolbox default constructor.

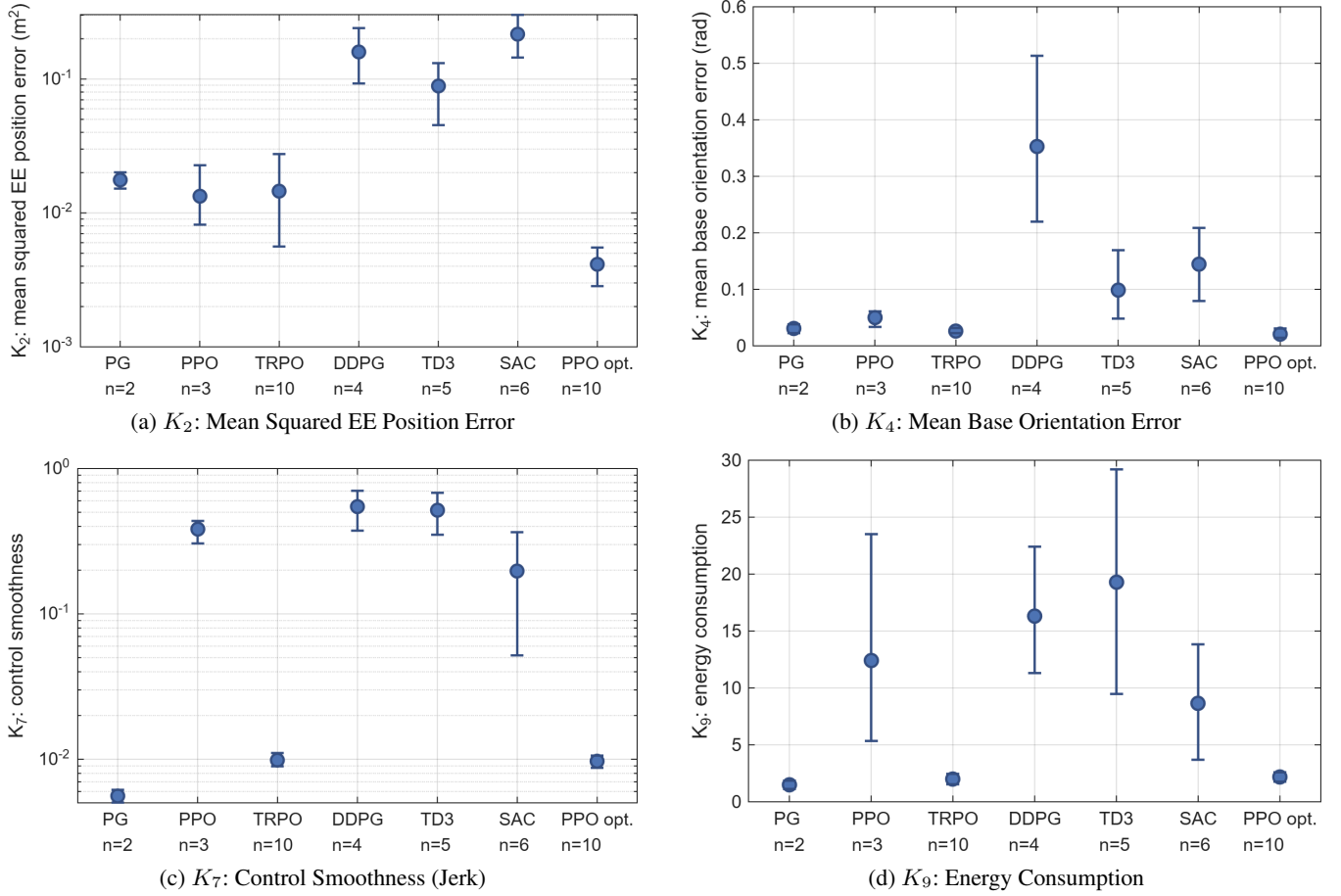


FIGURE 13: Mean and 95% bootstrap confidence interval of selected KPIs over the successful runs of each agent (2000 resamples). The number of successful runs is given below each agent. The intervals of agents with few successful runs are wide or undefined. The optimized PPO agent of Section VII is included for reference.

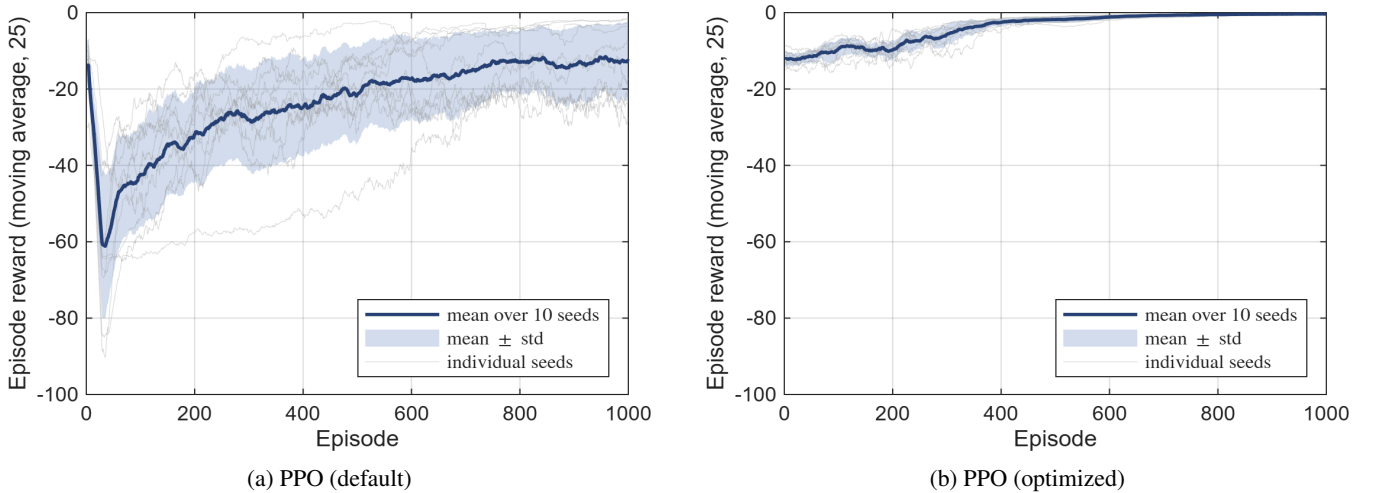


FIGURE 14: Episode rewards during training of the default and the optimized PPO agents over ten seeds. The thin lines show the moving average over 25 episodes of each seed, the thick line the mean over the seeds, and the band one standard deviation. Both panels use the same axis. (a) Default hyperparameters. (b) Optimized configuration.

Hyperparameter Tuning. The configuration was determined in a preliminary coordinate-wise grid search and is

kept fixed for all experiments reported here. The search used a single training run per candidate. It was carried out on an

TABLE 7: Success of the default agents over ten training runs each. A run is successful when none of its 31 evaluation episodes is terminated by a monitor. The abort rate is the share of terminated episodes among the 310 evaluation episodes of an agent. The share caused by a detected contact is listed separately, and the remaining terminations were caused by joint-limit violations. The last column gives the Holm-corrected p -value of Fisher's exact test on the number of successful runs against TRPO.

Agent	Successful runs	Abort rate	Aborts by contact	p vs. TRPO
TRPO	10/10	0.0%	–	–
SAC	6/10	40.0%	0.0%	0.087
TD3	5/10	48.4%	24.7%	0.065
DDPG	4/10	51.6%	22.5%	0.033
PPO	3/10	70.0%	14.3%	0.012
PG	2/10	80.0%	1.6%	0.0036

earlier version of the simulation model, before the reward weights, the termination rule, and the evaluation protocol of this work were fixed. The selected values are therefore used as a fixed configuration and are not claimed to be optimal for the final setup. Each hyperparameter was swept over a discrete set of candidate values while all others were held at their current best values. The configuration with the highest mean episodic reward over the final 50 of the $N_{\text{train}} = 1000$ training episodes was selected. In total, 16 training runs of 1000 episodes each were conducted during this tuning phase. The swept parameters, their candidate values, and the selected values are as follows:

- *Actor learning rate.* Candidates $\{1 \times 10^{-3}, 5 \times 10^{-4}\}$, selected 1×10^{-3} . Both candidates are an order of magnitude below the toolbox default of 10^{-2} , which reduces the size of the policy updates.
- *Critic learning rate.* Candidates $\{1 \times 10^{-3}, 5 \times 10^{-4}\}$, selected 5×10^{-4} , which gave the highest mean episodic reward in the search.
- *Entropy loss weight.* Candidates $\{1 \times 10^{-2}, 1 \times 10^{-3}\}$, selected 1×10^{-3} . A small entropy bonus is intended to prevent a premature convergence to a suboptimal deterministic policy while avoiding undirected exploration.
- *Experience horizon.* Candidates $\{512, 1024, 2048\}$, selected 1024. A moderate rollout horizon is a compromise between the trajectory information per update and the memory requirements and update latency.
- *Mini-batch size.* Candidates $\{128, 256\}$, selected 256. Larger mini-batches are expected to give more stable gradient estimates and thus a lower training variance.
- *Fixed settings.* The discount factor was set to $\gamma = 0.995$, the number of epochs per update to 10, and the gradient threshold of both optimizers to 1. The clip factor of 0.2 and the GAE factor of 0.95 remain at their default values.

In the tuning runs, the selected configuration gave the highest mean episodic reward. Compared with the default

configuration, the optimized agent uses learning rates that are ten to twenty times smaller, larger rollouts and mini-batches, more epochs per update, and gradient clipping. Together, these settings limit the size of each policy update.

A. CIRCULAR TRAJECTORY

Training Outcome. Fig. 14 shows the training curves of both configurations over ten seeds. The moving-average return of the default PPO agent dropped to about -61 within the first 35 episodes and then rose slowly to -12.7 after 1000 episodes, with a large spread between the seeds. The optimized configuration started at about -12 , exceeded -1 after 625 episodes, and reached -0.36 with a small spread between the seeds. For comparison, TRPO reached -0.81 after 1000 episodes.

Success and Performance. Table 10 compares the optimized PPO agent with the default PPO agent and with TRPO. The optimized configuration completed all ten runs without a terminated evaluation episode, compared with three runs for the default configuration (Fisher's exact test, $p = 0.003$). Over the successful runs, it reached a mean squared tracking error of $K_2 = 0.0041$, compared with 0.0134 for the default PPO agent and 0.0145 for TRPO. The base orientation error K_4 was 0.021, compared with 0.050 and 0.026. The optimized agent also produced smoother torque commands than the default PPO agent ($K_7, 0.383 \rightarrow 0.0097$) with a lower energy consumption ($K_9, 12.4 \rightarrow 2.16$). Its control smoothness and energy consumption were similar to those of TRPO.

Comparison With TRPO. On the seed level, only the difference in the base orientation error reached the 5% level (K_4 , Mann-Whitney, $p = 0.045$). The differences in the return K_1 , the tracking error K_2 , and the base angular velocity K_8 were not significant (p between 0.076 and 0.089). Given the number of KPIs tested, the single significant result should be interpreted with care. The optimized agent trained in 9.2 min per run, compared with 11.3 min for TRPO.

Fig. 15 shows the end-effector path of one evaluation episode for each PPO configuration. The default PPO agent follows the first quarter of the semicircle but oscillates around the reference in the second half, which is consistent with its large jerk K_7 . The optimized agent follows the reference without visible oscillation and stays slightly inside the circle near the end of the path. Fig. 16 shows the base orientation during the same episodes. The largest vector component of the quaternion was 0.028 for the default and 0.013 for the optimized agent.

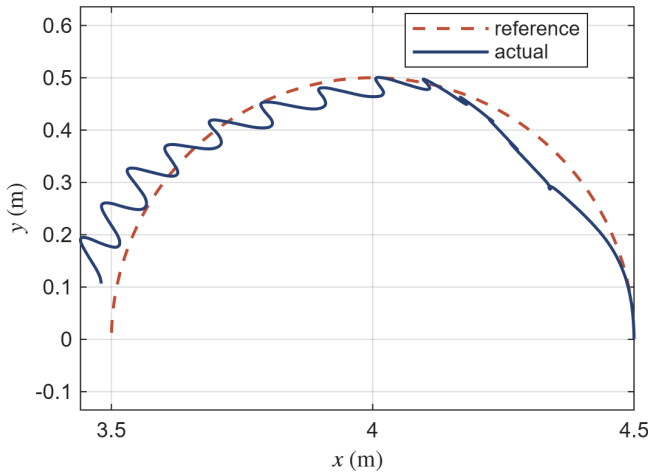
Under the tested conditions, the tuning removed the seed dependence of the default PPO agent. The tuned agent reached the tracking accuracy and base stability of TRPO with a shorter training time. Its lower tracking error compared with TRPO was not statistically significant with ten runs per configuration.

TABLE 8: Performance of the successful runs of the default agents. All values are the mean and the standard deviation over the successful runs of an agent. K_2 is the mean squared EE position error, K_4 the mean base orientation error, and T_2 the training time. The collision rate K_5 and the joint limit violation rate K_6 are zero for every successful run and are omitted.

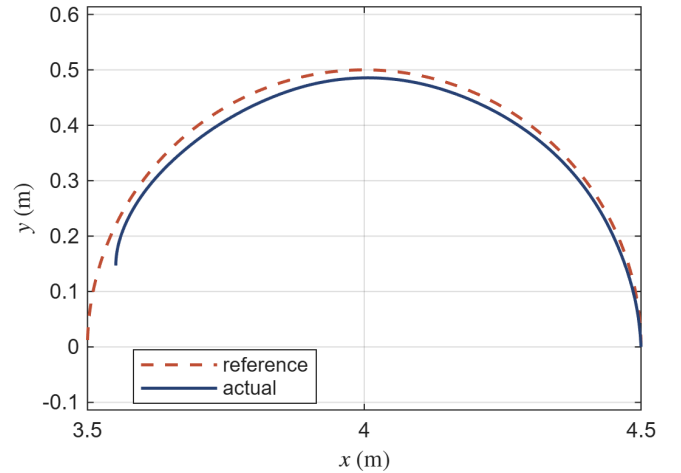
Agent	Successful runs	K_2	K_4	T_2 (min)
TRPO	10	0.0145 ± 0.0196	0.026 ± 0.009	11.3 ± 0.7
SAC	6	0.2168 ± 0.1098	0.145 ± 0.092	35.2 ± 0.4
TD3	5	0.0886 ± 0.0560	0.098 ± 0.080	29.1 ± 1.4
DDPG	4	0.1599 ± 0.0919	0.353 ± 0.171	24.3 ± 0.2
PPO	3	0.0134 ± 0.0081	0.050 ± 0.014	8.4 ± 0.7
PG	2	0.0177 ± 0.0036	0.030 ± 0.011	6.9 ± 1.3

TABLE 9: Seed-level comparison of the successful runs of the default agents with those of TRPO. Each cell gives the median difference $\Delta\hat{x}$ (agent minus TRPO) with the Holm-corrected significance of a two-sided Mann–Whitney U test as superscript ($***p < 0.001$, $**p < 0.01$, $*p < 0.05$, $n.s.p \geq 0.05$). For K_1 , a negative difference means a lower return than TRPO. For the other KPIs, a positive difference means a larger error, jerk, angular velocity, or energy than TRPO. PPO and PG are not tested because they have fewer than four successful runs.

Agent	n	K_1	K_2	K_3	K_4	K_7	K_8	K_9
SAC	6	-8.08^{***}	$+0.180^{***}$	$+0.779^{***}$	$+0.134^*$	$+0.125^{n.s.}$	$+0.027^*$	$+7.06^{n.s.}$
TD3	5	-3.07^{**}	$+0.085^{**}$	$+0.387^{**}$	$+0.046^{**}$	$+0.532^{**}$	$+0.027^{**}$	$+18.3^{**}$
DDPG	4	-7.81^{**}	$+0.128^{**}$	$+0.770^{**}$	$+0.299^{**}$	$+0.562^{**}$	$+0.062^{**}$	$+13.1^{**}$



(a) PPO (default)



(b) PPO (optimized)

FIGURE 15: Reference and actual end-effector path in the XY plane for one evaluation episode from the nominal initial state. Each panel shows the successful run with the median K_2 of its configuration, seed 2 for the default and seed 0 for the optimized PPO agent.

B. ABLATION STUDY OF REWARD COMPONENTS

An ablation study quantifies the contribution of each term of the reward function in (1). The optimized PPO agent was re-trained on the circular trajectory with one reward component disabled at a time, with five seeds ($s = 0, \dots, 4$) per variant. Each of the seven weights in the cost term c_t was set to zero in turn while the remaining six were kept at their baseline values ($W_p = 150$, $W_v = 25$, $W_{ori} = 200$, $W_{\omega_b} = 8$, $W_{v_b} = 2$, $W_u = 0.02$, $W_d = 0.06$). In addition, the two positive reward terms were disabled by setting $k_p = 0$ and $k_b = 0$, which removes r_{prog} and r_{bonus} . All other settings

of training and evaluation were identical to those of the optimized PPO agent. All variants were evaluated with the nominal reward weights, so that the return K_1 is comparable across the variants. The reference row contains the optimized PPO agent of Table 10 with the same five seeds. The results are summarized in Table 11.

Dominant Weights (W_p , W_{ori}). Removing the end-effector position weight W_p had the largest effect on tracking. The mean squared tracking error K_2 rose from 0.0050 to 0.259, about 52 times the reference value, and the return K_1 dropped from -0.20 to -6.79 . The base orientation

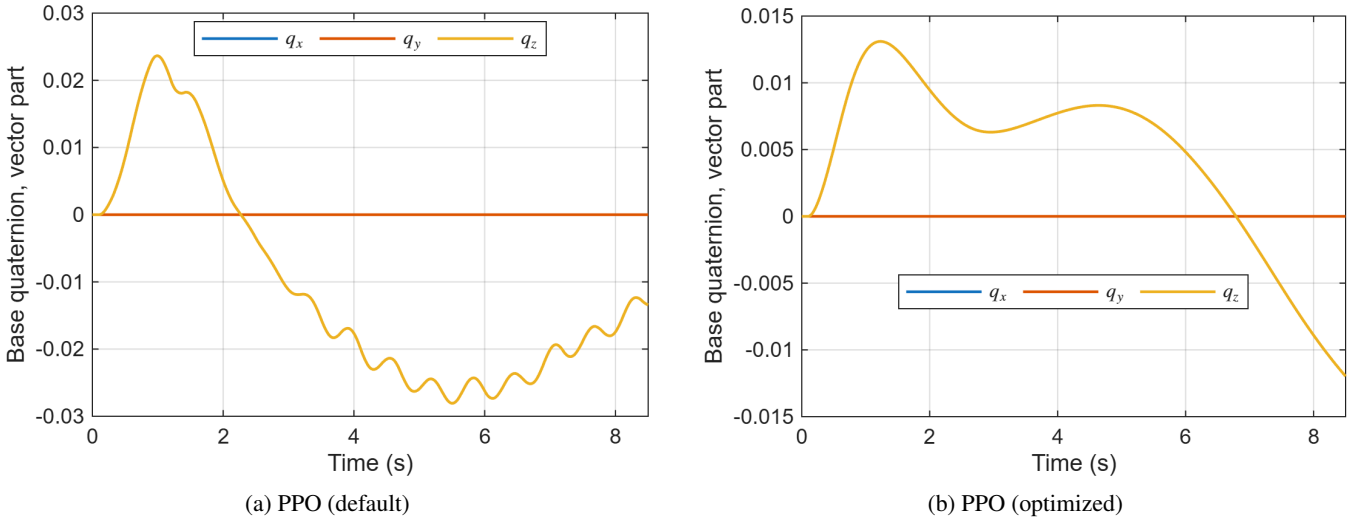


FIGURE 16: Vector part of the base orientation quaternion during the episodes of Fig. 15. The base rotates only about the z -axis. The scalar part stays above 0.9996 in both episodes.

TABLE 10: Default PPO agent, optimized PPO agent, and TRPO on the circular path. The first row gives the number of successful runs out of ten. All other values are the mean and the standard deviation over the successful runs. K_5 and K_6 are zero for every successful run and are omitted.

Metric (KPI)	PPO (default)	PPO (optimized)	TRPO
Successful runs	3/10	10/10	10/10
Mean Episodic Return (K_1)	-0.72 ± 0.12	-0.15 ± 0.12	-0.44 ± 0.55
Mean Squared EE Position Error (K_2)	0.0134 ± 0.0081	0.0041 ± 0.0023	0.0145 ± 0.0196
Maximum EE Position Error (K_3)	0.215 ± 0.040	0.168 ± 0.020	0.229 ± 0.095
Mean Base Orientation Error (K_4)	0.050 ± 0.014	0.021 ± 0.016	0.026 ± 0.009
Control Smoothness (K_7 , Jerk)	0.383 ± 0.068	0.0097 ± 0.0017	0.0098 ± 0.0017
Mean Base Angular Velocity (K_8)	0.031 ± 0.007	0.012 ± 0.005	0.017 ± 0.008
Energy Consumption (K_9)	12.4 ± 9.7	2.16 ± 0.69	2.00 ± 0.78
Reward Std. (Late Training, T_1)	1.23 ± 0.58	0.098 ± 0.027	0.26 ± 0.18
Training Time (T_2 , min)	8.4 ± 0.7	9.2 ± 0.6	11.3 ± 0.7

error decreased at the same time (K_4 , $0.025 \rightarrow 0.014$), which suggests that the agent then mainly stabilized the base. The low energy consumption of this variant ($K_9 = 0.56$) reflects the reduced arm motion and does not indicate a higher efficiency. Removing the orientation weight W_{ori} had the opposite effect. K_4 rose about fourfold to 0.104, one of the five runs failed, and K_2 decreased to 0.0030. W_p and W_{ori} therefore balance end-effector tracking against base stabilization.

Remaining Terms. Disabling any of the other five cost weights or one of the two positive reward terms changed the KPIs little. K_2 stayed between 0.0036 and 0.0063 and K_4 between 0.015 and 0.035, and all runs succeeded. These differences are small compared with the spread between the seeds of the reference. In this setting, the low-weight terms and the positive reward terms therefore had no clear effect on the final performance. Their effect on the course of training was not analyzed.

Summary. The ablation indicates that W_p and W_{ori} are the two dominant components of the reward. The optimized

agent solved the task without any one of the other seven terms. This suggests that the results of the optimized PPO agent do not depend on a fine balance of the low-weight terms.

C. SENSITIVITY ANALYSIS OF REWARD WEIGHTS

The ablation study in Section VII-B identifies the dominant reward components. A complementary question is how sensitive the results are to a moderate miscalibration of the weights and whether the ranking of the algorithms in Section VI depends on them. Two analyses address this question. First, four cost weights of the optimized PPO agent were multiplied by 0.5 and by 2 one at a time. These are the two dominant weights W_p and W_{ori} and the secondary weights W_v and W_{ω_b} . Second, TRPO, SAC, and the default PPO agent were retrained with W_p and W_{ori} multiplied by 0.5 and by 2. All variants were trained with five seeds ($s = 0, \dots, 4$) and evaluated with the nominal reward weights. The reference rows contain the same agents and seeds of Section VI and Section VII.

TABLE 11: Ablation study of the reward components on the circular trajectory. Each row corresponds to the optimized PPO agent retrained with five seeds and one reward term disabled. The values are the mean and the standard deviation over the successful runs, evaluated with the nominal reward weights. K_5 and K_6 are zero for every successful run and are omitted.

Variant	Successful runs	K_1	K_2	K_4	K_7	K_9
Nominal weights	5/5	-0.20 ± 0.13	0.0050 ± 0.0016	0.025 ± 0.021	0.0101 ± 0.0013	2.22 ± 0.85
$W_p = 0$	5/5	-6.79 ± 1.41	0.2592 ± 0.0552	0.014 ± 0.005	0.0046 ± 0.0007	0.56 ± 0.30
$W_v = 0$	5/5	-0.26 ± 0.18	0.0054 ± 0.0030	0.033 ± 0.025	0.0088 ± 0.0013	1.99 ± 0.42
$W_{ori} = 0$	4/5	-0.83 ± 0.51	0.0030 ± 0.0015	0.104 ± 0.053	0.0095 ± 0.0018	2.01 ± 0.29
$W_{\omega_b} = 0$	5/5	-0.14 ± 0.10	0.0041 ± 0.0026	0.022 ± 0.007	0.0086 ± 0.0009	2.00 ± 0.62
$W_{v_b} = 0$	5/5	-0.16 ± 0.08	0.0048 ± 0.0022	0.017 ± 0.014	0.0088 ± 0.0003	2.12 ± 0.55
$W_u = 0$	5/5	-0.14 ± 0.10	0.0042 ± 0.0030	0.019 ± 0.010	0.0094 ± 0.0014	1.94 ± 0.48
$W_d = 0$	5/5	-0.27 ± 0.19	0.0063 ± 0.0042	0.035 ± 0.025	0.0100 ± 0.0016	2.45 ± 0.46
$k_p = 0$ (r_{prog} off)	5/5	-0.11 ± 0.06	0.0036 ± 0.0020	0.015 ± 0.006	0.0084 ± 0.0014	2.16 ± 0.87
$k_b = 0$ (r_{bonus} off)	5/5	-0.23 ± 0.09	0.0057 ± 0.0042	0.031 ± 0.018	0.0095 ± 0.0011	2.03 ± 0.43

TABLE 12: One-at-a-time sensitivity analysis of four reward weights of the optimized PPO agent on the circular trajectory. Each row corresponds to the agent retrained with five seeds and one weight multiplied by 0.5 or 2. The values are the mean and the standard deviation over the successful runs, evaluated with the nominal reward weights.

Variant	Successful runs	K_1	K_2	K_4	K_7	K_9
Nominal weights	5/5	-0.20 ± 0.13	0.0050 ± 0.0016	0.025 ± 0.021	0.0101 ± 0.0013	2.22 ± 0.85
$W_p \times 0.5$	5/5	-0.33 ± 0.14	0.0114 ± 0.0049	0.018 ± 0.007	0.0082 ± 0.0004	1.83 ± 0.53
$W_p \times 2$	5/5	-0.55 ± 0.69	0.0063 ± 0.0043	0.070 ± 0.060	0.0093 ± 0.0015	2.81 ± 1.00
$W_v \times 0.5$	5/5	-0.09 ± 0.03	0.0026 ± 0.0010	0.016 ± 0.003	0.0089 ± 0.0006	2.41 ± 0.35
$W_v \times 2$	5/5	-0.19 ± 0.17	0.0058 ± 0.0051	0.021 ± 0.012	0.0119 ± 0.0025	2.32 ± 0.67
$W_{ori} \times 0.5$	5/5	-0.18 ± 0.11	0.0044 ± 0.0030	0.028 ± 0.013	0.0092 ± 0.0006	2.24 ± 0.34
$W_{ori} \times 2$	5/5	-0.20 ± 0.19	0.0072 ± 0.0072	0.012 ± 0.005	0.0085 ± 0.0013	2.10 ± 0.29
$W_{\omega_b} \times 0.5$	5/5	-0.13 ± 0.06	0.0038 ± 0.0015	0.018 ± 0.006	0.0100 ± 0.0007	2.21 ± 0.38
$W_{\omega_b} \times 2$	5/5	-0.16 ± 0.10	0.0051 ± 0.0037	0.015 ± 0.006	0.0079 ± 0.0011	1.89 ± 0.33

TABLE 13: Success and tracking error of TRPO, the default PPO agent, and SAC under changed reward weights on the circular trajectory. Each cell gives the number of successful runs out of five and the mean K_2 over the successful runs.

Variant	TRPO	PPO (default)	SAC
Nominal weights	5/5, 0.0078	2/5, 0.0160	3/5, 0.219
$W_{ori} \times 0.5$	5/5, 0.0045	1/5, 0.067	2/5, 0.175
$W_{ori} \times 2$	5/5, 0.0307	1/5, 0.061	3/5, 0.246
$W_p \times 0.5$	5/5, 0.0180	0/5, –	2/5, 0.448
$W_p \times 2$	5/5, 0.0102	2/5, 0.072	1/5, 0.146

Optimized PPO Agent. All 40 runs of the eight perturbed configurations succeeded (Table 12). The tracking error K_2 ranged from 0.0026 to 0.0114, compared with 0.0050 at the nominal weights, and the base orientation error K_4 ranged from 0.012 to 0.070. The largest changes occurred for the two dominant weights. Halving W_p increased K_2 to 0.0114, and doubling W_p increased K_4 to 0.070. Doubling W_{ori} reduced K_4 to 0.012 and increased K_2 to 0.0072. These shifts reflect the trade-off between tracking and base stabilization identified in the ablation. The perturbations of W_v and W_{ω_b} led to smaller changes, with K_2 between 0.0026 and 0.0058.

Ranking of the Algorithms. Table 13 shows TRPO, the default PPO agent, and SAC under the four weight variants. TRPO succeeded in all 20 runs of the variants and had the

lowest mean tracking error in every variant. For TRPO, doubling W_{ori} increased the mean K_2 to 0.0307, mainly because of one weak run with $K_2 = 0.105$, and the median was 0.0056. The default PPO agent succeeded in zero to two of five runs per variant, and SAC in one to three. The successful runs of SAC tracked the path with K_2 between 0.146 and 0.448, roughly an order of magnitude above TRPO. In all tested variants, the order of the three algorithms by tracking error was the same as with the nominal weights.

Summary. Within the tested range from 0.5 to 2 times the nominal values, the optimized PPO agent succeeded in every run, and the reward weights mainly shifted the balance between tracking and base stabilization. TRPO succeeded in every run, and the order of TRPO, PPO, and SAC by tracking error did not change in any tested variant. The analysis covers changes of single weights and three algorithms. Interactions between weights and the behavior of the remaining algorithms were not tested.

D. ROBUSTNESS EVALUATION UNDER NON-IDEAL CONDITIONS

The stress tests evaluate the trained policies beyond the nominal simulation assumptions. All 70 policies of the benchmark were evaluated without retraining on the same 31 initial states under six non-ideal conditions. Sensor noise was added to the observations seen by the agent as zero-mean Gaussian noise

with a standard deviation of 0.005 in the unit of each channel. The plant and the reward used the undisturbed signals. The torque limit of the plant was reduced to $0.75 \tau_{\max}$. Because this limit had almost no effect on TRPO and the optimized PPO agent, a stronger reduction to $0.3 \tau_{\max}$ was added. An external disturbance was applied as an additive joint torque after the torque-saturation block, with $\tau_{\text{dist},1} = \tau_{\text{dist},2} = 2 \text{ N m}$ for $2.0 \leq t < 2.5 \text{ s}$. Parameter uncertainty was modeled by multiplying all masses, inertias, and joint damping coefficients by 1.5. The combined case applied the noise, the torque limit of $0.75 \tau_{\max}$, the disturbance, and the parameter change at the same time.

TRPO and Optimized PPO Agent. Both agents completed all runs without a terminated episode under every tested condition (Table 14). Sensor noise and the torque limit of $0.75 \tau_{\max}$ changed their tracking and base orientation errors by less than 1%. The noise increased the jerk K_7 of both agents by about two thirds, from 0.0097 to 0.0161 for the optimized PPO agent, likely because the noise enters the torque commands directly. The torque limit of $0.3 \tau_{\max}$ increased the tracking error of the optimized PPO agent by 18% (0.00415 to 0.00489) and changed that of TRPO by 2%. The external disturbance mainly affected the base. K_4 rose from 0.021 to 0.055 for the optimized PPO agent and from 0.026 to 0.066 for TRPO, while K_2 stayed at 0.00506 and 0.01289. The parameter change had the largest effect on tracking. K_2 rose to 0.01421 for the optimized PPO agent and to 0.02847 for TRPO, about three and two times the nominal values. The combined case showed the same increase in K_2 together with the increase in K_4 caused by the disturbance. Under every condition, the optimized PPO agent kept a lower mean tracking error than TRPO.

Other Agents. Under the torque limit of $0.3 \tau_{\max}$, more runs of the other default agents succeeded. TD3 succeeded in nine, SAC in eight, DDPG in seven, and the default PPO agent in five of ten runs, compared with five, six, four, and three in the nominal case. This suggests that the lower limit prevented part of the fast motions that otherwise ended at a joint limit. The tracking error of TD3, DDPG, and SAC remained high in these runs (K_2 between 0.139 and 0.248). Under the external disturbance, fewer runs of SAC, TD3, and DDPG succeeded (five, four, and two runs).

Scope. The monitored properties were not violated by TRPO and the optimized PPO agent in any stress test. Each perturbation was tested at a single level, except for the torque limit, and the noise model assumes independent Gaussian noise per channel and step. These stress tests therefore do not replace a systematic robustness characterization.

E. PIECEWISE-LINEAR TRAJECTORY

The previous evaluations consider tracking of a smooth circular end-effector reference. To assess whether the results carry over to a different reference geometry, the optimized PPO agent, the default PPO agent, and TRPO were retrained with ten seeds on a piecewise-linear trajectory. The hyperparameters and the reward weights were the same as for

the circular path. The experiment therefore tests whether the configurations carry over to another path, not whether a policy trained on the circle generalizes. This case is relevant for on-orbit manipulation tasks in which approach and retreat motions are planned as straight-line segments rather than continuous curves.

Trajectory Definition. The linear reference uses the same circle parameters (center and radius) as the circular benchmark. The end-effector is commanded to move along two straight segments that connect three points on that circle, $P_0 = [c_x + r, c_y, c_z]$, $P_1 = [c_x, c_y + r, c_z]$, and $P_2 = [c_x - r, c_y, c_z]$. Using a period T and a switching time $t_1 = T/2$, the piecewise-linear reference position $p_{\text{ref}}(t)$ is

$$p_{\text{ref}}(t) = \begin{cases} P_0 + \frac{t}{t_1} (P_1 - P_0), & 0 \leq t \leq t_1, \\ P_1 + \frac{t-t_1}{T-t_1} (P_2 - P_1), & t_1 < t \leq T. \end{cases} \quad (6)$$

The reference is sampled at $T_s = 0.01 \text{ s}$, and the agents act every $\Delta t = 0.1 \text{ s}$ as in the other experiments.

Results. Table 15 summarizes the results on the piecewise-linear reference. The optimized PPO agent and TRPO completed all ten runs without a terminated episode, whereas the default PPO agent succeeded in only one run (Fisher's exact test against TRPO, $p < 0.001$). Over the successful runs, the optimized PPO agent reached a mean squared tracking error of $K_2 = 0.0059$, compared with 0.0125 for TRPO. On the seed level, the optimized PPO agent had a significantly higher return and lower mean and maximum tracking errors than TRPO (K_1 , K_2 , and K_3 , Mann-Whitney, $p \leq 0.038$). The difference in the base orientation error was not significant (K_4 , 0.019 compared with 0.029, $p = 0.054$).

Fig. 17 shows the end-effector path of one episode for the default and the optimized PPO agent. The only successful run of the default agent lags behind the reference, oscillates along both segments, and ends far from the final point. The optimized agent approaches the first segment on a curved path, cuts the corner, and then follows the second segment closely. Fig. 18 shows the corresponding base orientations. The largest vector component of the quaternion was 0.121 for the default and 0.039 for the optimized agent.

Under the tested conditions, the configurations of TRPO and the optimized PPO agent carried over to the piecewise-linear path without changes, and the optimized agent tracked this path more accurately than TRPO. The default PPO agent succeeded in only one of ten runs.

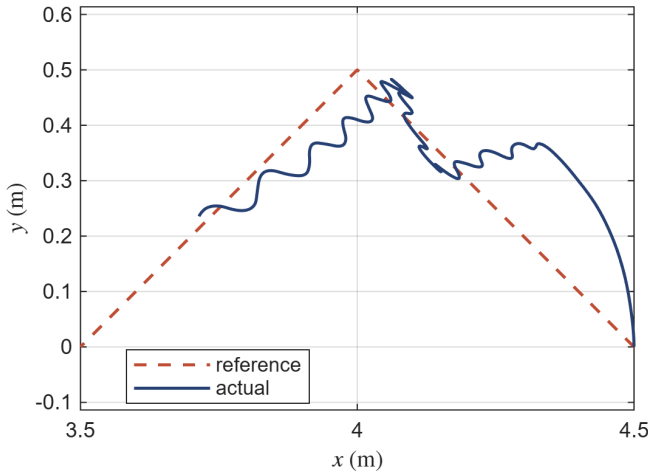
The following section discusses the implications, limitations, and future steps toward deployment on physical testbeds for space robotics.

VIII. DISCUSSION AND CONCLUSION

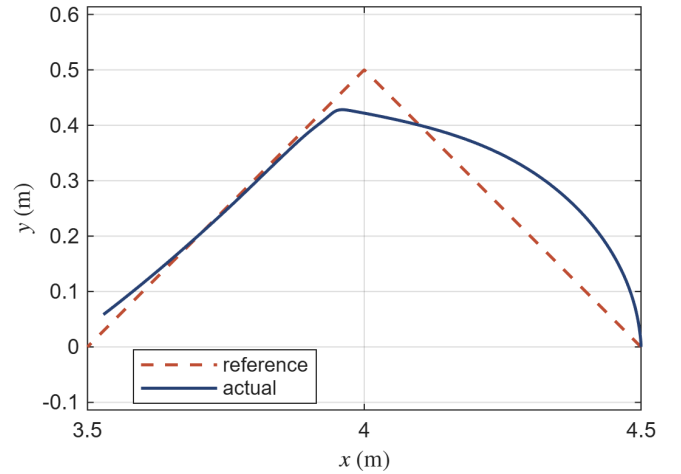
This work investigated continuous-action deep RL for the control of a free-floating space robotic arm, with a focus on control performance and the monitored safety properties. The results show that an RL-based controller can perform the considered manipulation task in the simulated space environment under the stated assumptions. With the toolbox

TABLE 14: Stress tests of the trained policies on the circular trajectory, evaluated without retraining. TRPO and the optimized PPO agent completed all ten runs under every condition. For these two agents, the table gives the mean and the standard deviation of K_2 and K_4 over the ten runs. For the other default agents, it gives the number of successful runs out of ten.

Condition	TRPO		PPO (optimized)		Successful runs				
	K_2	K_4	K_2	K_4	SAC	TD3	DDPG	PPO	PG
Nominal	0.01454 ± 0.0196	0.026 ± 0.009	0.00415 ± 0.0023	0.021 ± 0.016	6	5	4	3	2
Sensor noise	0.01455 ± 0.0196	0.026 ± 0.009	0.00415 ± 0.0023	0.021 ± 0.016	6	5	3	3	2
Torque limit $0.75 \tau_{\max}$	0.01454 ± 0.0196	0.026 ± 0.009	0.00415 ± 0.0023	0.021 ± 0.016	6	6	5	3	2
Torque limit $0.3 \tau_{\max}$	0.01488 ± 0.0197	0.026 ± 0.010	0.00489 ± 0.0031	0.021 ± 0.015	8	9	7	5	2
External disturbance	0.01289 ± 0.0145	0.066 ± 0.024	0.00506 ± 0.0028	0.055 ± 0.023	5	4	2	3	2
Parameters $\times 1.5$	0.02847 ± 0.0267	0.022 ± 0.008	0.01421 ± 0.0061	0.019 ± 0.011	7	7	5	3	2
Combined	0.02790 ± 0.0228	0.058 ± 0.021	0.01434 ± 0.0060	0.049 ± 0.021	6	7	5	4	2



(a) PPO (default)



(b) PPO (optimized)

FIGURE 17: Reference and actual end-effector path on the piecewise-linear reference for one evaluation episode from the nominal initial state. The panels show the only successful run of the default PPO agent (seed 9) and the run of the optimized PPO agent with the median K_2 (seed 5).

default hyperparameters, the agents differed more in how often training succeeded than in the performance of their successful runs. TRPO completed all ten runs, whereas the other default agents terminated 40.0% to 80.0% of their evaluation episodes. The tuned PPO agent also completed all runs. It reached a mean squared tracking error of $K_2 = 0.0041$, compared with 0.0145 for TRPO and 0.0134 for the successful runs of the default PPO agent. Its advantage over TRPO was not statistically significant, but it trained faster. No violations of the monitored safety properties were recorded for TRPO and the tuned PPO agent in any evaluation episode [1], [22].

An additional evaluation on a piecewise-linear end-effector reference tested whether the configurations carry over to a different path. With the same hyperparameters and reward weights, the optimized PPO agent and TRPO again completed all ten runs, and the optimized agent tracked the path significantly more accurately than TRPO (K_2 , 0.0059 compared with 0.0125). The default PPO agent succeeded in only one of ten runs. The ablation and the sensitivity analysis indicate that W_p and W_{ori} dominate the reward design. The

order of TRPO, PPO, and SAC by tracking error did not change when these weights were halved or doubled. In the stress tests, TRPO and the optimized PPO agent completed all runs under sensor noise, reduced torque limits, an external joint torque, and a 50% change of the model parameters. The parameter change increased their tracking errors by a factor of two to three. These results are limited to a few perturbation levels and two reference paths [44].

Practical Implications. The results of TRPO and the tuned PPO agent indicate that policy-gradient methods with constrained updates are suitable candidates for continuous control tasks with strong dynamic coupling and safety constraints. For PPO, this held only with a smaller learning rate than the toolbox default. With the default learning rate of 10^{-2} , the clipped surrogate objective of PPO did not prevent the seed dependence observed in Section VI. The runtime monitors (collision monitor and joint-limit monitor) supervise the training. They terminate episodes in which the policy would violate a monitored constraint and apply a terminal penalty [20], [24], [25]. Combined with the penalty-

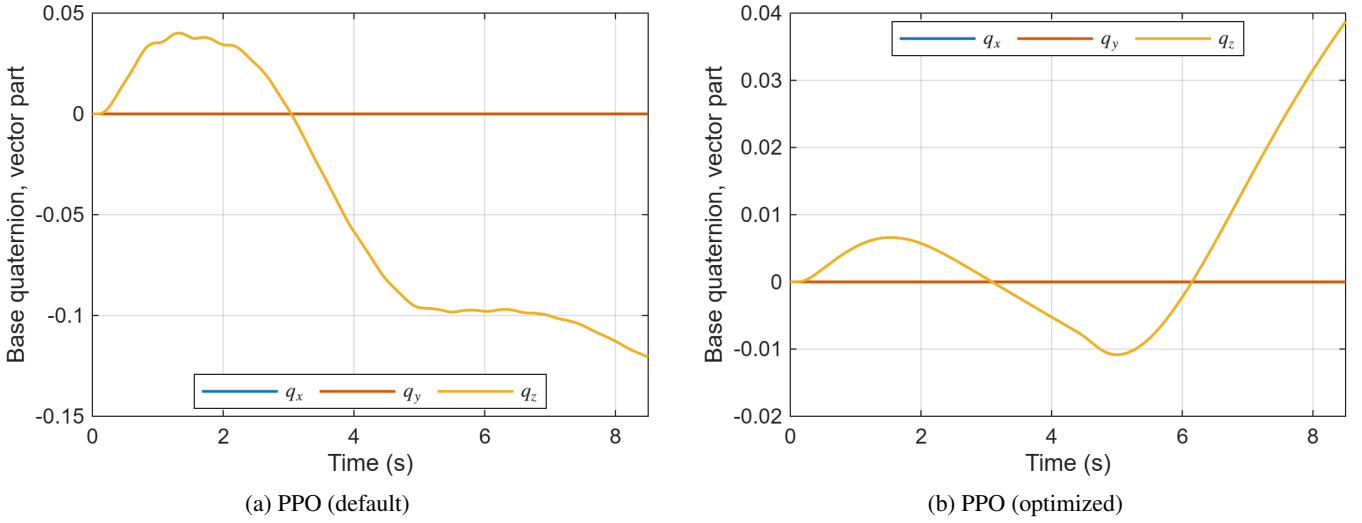


FIGURE 18: Vector part of the base orientation quaternion during the episodes of Fig. 17. The base rotates only about the z -axis. The scalar part stays above 0.992 for the default and above 0.999 for the optimized agent.

TABLE 15: Default PPO agent, optimized PPO agent, and TRPO on the piecewise-linear trajectory. The first row gives the number of successful runs out of ten. All other values are the mean and the standard deviation over the successful runs, which is a single run for the default PPO agent. K_5 and K_6 are zero for every successful run and are omitted.

Metric (KPI)	PPO (default)	PPO (optimized)	TRPO
Successful runs	1/10	10/10	10/10
Mean Episodic Return (K_1)	-2.43	-0.19 ± 0.15	-0.41 ± 0.37
Mean Squared EE Position Error (K_2)	0.0528	0.0059 ± 0.0053	0.0125 ± 0.0105
Maximum EE Position Error (K_3)	0.415	0.150 ± 0.013	0.194 ± 0.055
Mean Base Orientation Error (K_4)	0.129	0.019 ± 0.011	0.029 ± 0.023
Control Smoothness (K_7 , Jerk)	0.402	0.0117 ± 0.0019	0.0105 ± 0.0029
Mean Base Angular Velocity (K_8)	0.048	0.013 ± 0.006	0.020 ± 0.010
Energy Consumption (K_9)	11.8	3.42 ± 1.85	2.49 ± 1.26
Reward Std. (Late Training, T_1)	3.55	0.12 ± 0.11	0.94 ± 1.37
Training Time (T_2 , min)	6.08	7.30 ± 0.04	8.60 ± 0.11

driven reward, this led to final policies of TRPO and the tuned PPO agent that did not violate the monitored constraints in the reported evaluation. The present simulation study does not establish that this behavior transfers to a physical system. It suggests only that safety checks embedded in the training simulator may remain useful as runtime watchdogs, provided that the deployed system stays within the simulator's fidelity range, which remains to be validated [24], [31].

Limitations and Future Work. The study has several limitations. First, a sim-to-real gap remains. Beyond the idealizing assumptions stated in Section III (no sensor noise, no latency, rigid-body model, no external disturbances), real deployment introduces three further challenges. The first is a domain shift due to inaccurate mass and inertia parameters, joint friction, and link flexibility, which the URDF model does not capture. The second concerns onboard compute constraints. The learned policy must run in real time on embedded space-qualified hardware, typically ARM-based processors with limited floating-point throughput, whereas training was performed on a workstation CPU. The third

is sensor-actuator latency. Delays between state measurement and torque application can destabilize a policy that was trained without latency. Mitigating these effects would require domain randomization over simulation parameters, latency injection during training, and ultimately HIL validation [24], [44]. Second, the RL agent exhibits hyperparameter sensitivity. The tuned PPO configuration is task-specific and may require re-tuning for different trajectories or mission scenarios. It was selected in an earlier grid search with a single run per candidate. The other algorithms were compared only with their toolbox default hyperparameters, and tuning them might change their ranking. Third, the evaluation scope is limited. Each configuration was trained with ten seeds and evaluated on 31 initial states within $\pm 1^\circ$ of the nominal pose, for a single robot model and two reference paths. Ten runs per configuration allow only coarse estimates of the success rate, and the KPIs of agents with few successful runs rest on two to four runs. The stress tests cover a small set of perturbations. The reported ranking and improvements are therefore conditional on the tested setup. Fourth, the state representation

was manually designed. Learned representations or recurrent architectures (LSTM) might improve performance under partial observability. Fifth, the safety monitoring covers selected properties (collision events, joint limits, torque bound, and momentum consistency). It does not constitute a formal proof of closed-loop stability, task completion, or collision avoidance for the full nonlinear model, nor an exhaustive formal verification of the mission [23], [24]. Finally, no hardware testing was performed. Real-time computation limits, communication delays, and physical interactions remain to be validated on physical testbeds.

Future work could address these limitations in four directions. The first is domain randomization over simulation parameters and latency injection to improve the sim-to-real transfer [44] (see Fig. 19). The second is the extension to more complex tasks such as dual-arm coordination, on-orbit assembly, or active debris removal. The third is the use of recurrent policies, attention mechanisms, and safe RL algorithms with explicit constraint handling via Lagrange multipliers or barrier functions [24], [25], [31], [32]. The fourth is HIL validation on the authors' testbed for In-Space Robotized Services (IROS) with virtually simulated micro-gravity and predicted energy consumption (see Fig. 19).

Conclusion. This work has presented a framework that combines deep reinforcement learning, multibody simulation, and formal-methods-based safety monitoring for Cartesian path tracking of a free-floating space robot. Six continuous-action RL algorithms were compared under unified conditions with the toolbox default hyperparameters and ten seeds each. For the considered simulated task and evaluation setup, TRPO was the only default agent that completed all runs without a safety abort. A tuned PPO configuration also completed all runs and reached the lowest tracking error with a shorter training time than TRPO, although this difference was not statistically significant. Runtime monitoring of selected safety properties in the training loop, together with an offline torque-bound check, detected and terminated the violations of the other agents. These simulation results, obtained under the stated idealizing assumptions, do not establish formal safety or operational readiness, but they support further investigation of deep RL with runtime safety monitoring for space manipulation tasks.

The results also illustrate the influence of algorithm choice and hyperparameter tuning on the learning performance for the considered task. They are consistent with the view that no single RL algorithm is uniformly best across applications. Under unified conditions, the choice of algorithm and hyperparameters visibly affected the achieved tracking accuracy and base stability and, above all, how often training succeeded. Systematic benchmarking of candidate algorithms over several seeds is therefore a practical step in the selection of an appropriate RL controller for a given task.

Several open problems remain for real-world deployment, but the proposed framework provides a basis for further investigation. Possible application areas include satellite servicing, on-orbit assembly, and active debris removal, in

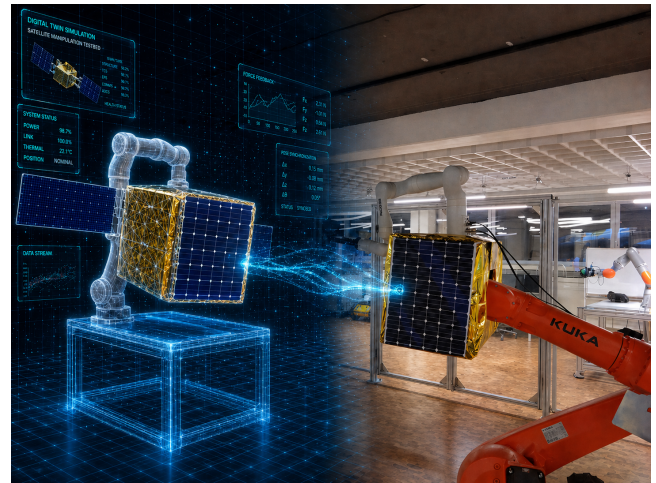


FIGURE 19: Gradual hybrid testbeds that integrate AI-driven digital twins (left) with physical robotic systems (right). Such testbeds allow policy optimization and domain-randomization strategies for sim-to-real transfer to be studied. Illustration created using a picture of the physical setup at Frankfurt University of Applied Sciences and with the help of GPT-Image-2.

which learning-based controllers could adapt to uncertain dynamics within predefined operational limits. One possible architecture for such applications combines an RL policy with a separate runtime layer that monitors the commands and overrides those that violate the specified constraints. In this combination, the learned policy provides adaptability, while the monitoring layer detects and intercepts violations of the specified properties [1], [25].

This work is one step in this direction. Future research at the intersection of robotics, reinforcement learning, and formal methods is required to extend the present results to physical deployment and to broader mission scenarios.

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